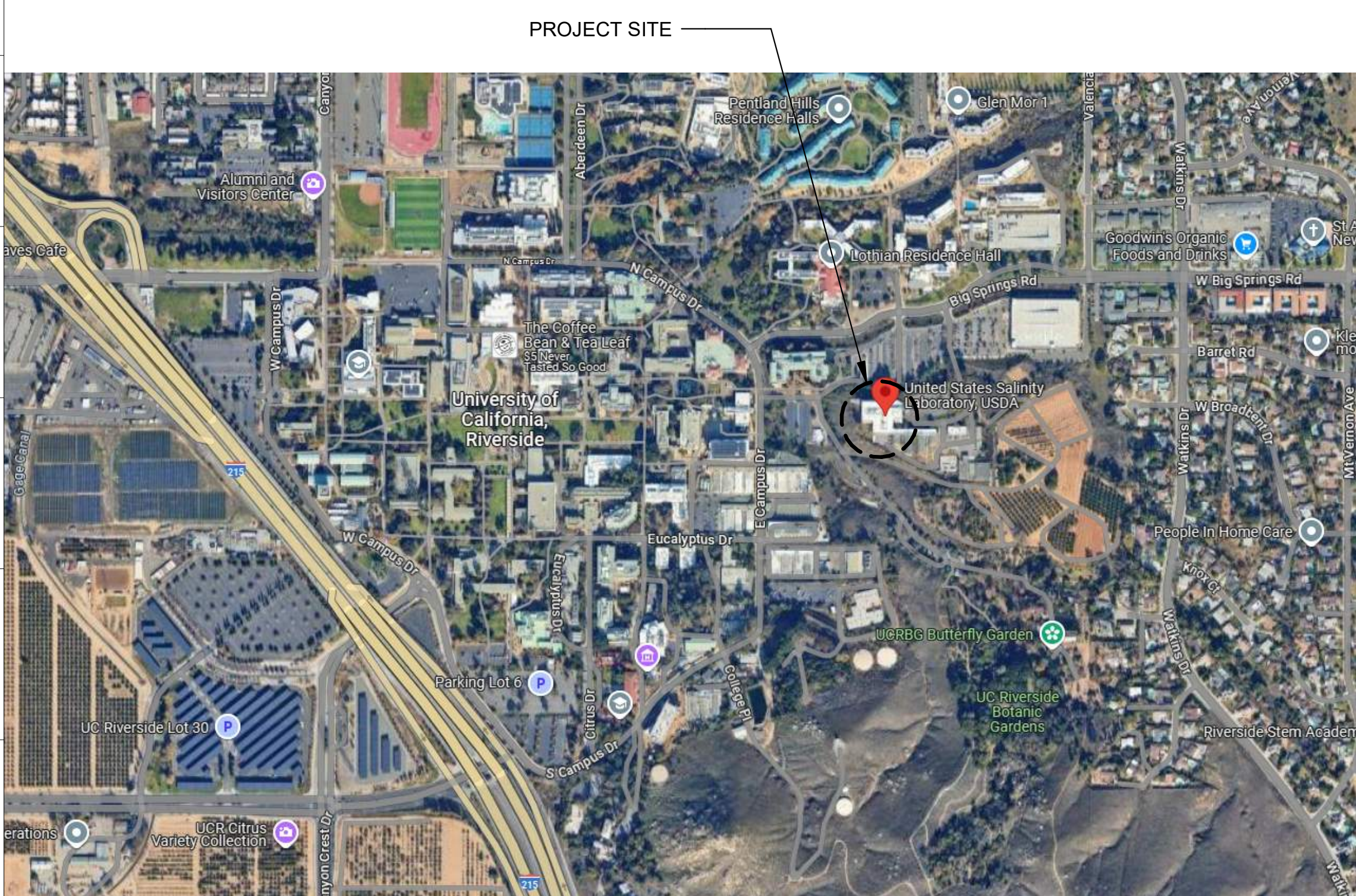




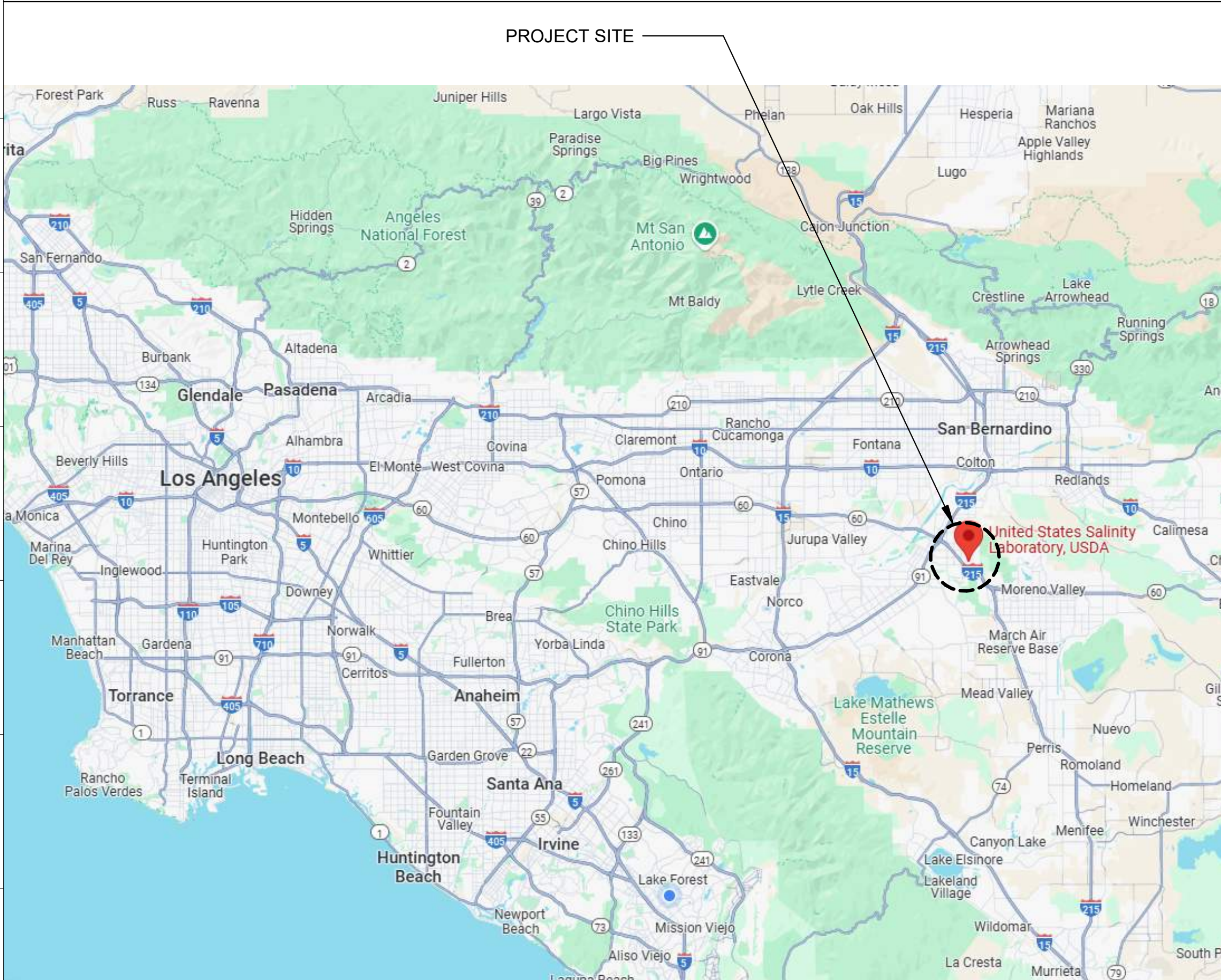
US SALINITY LABORATORY

UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA

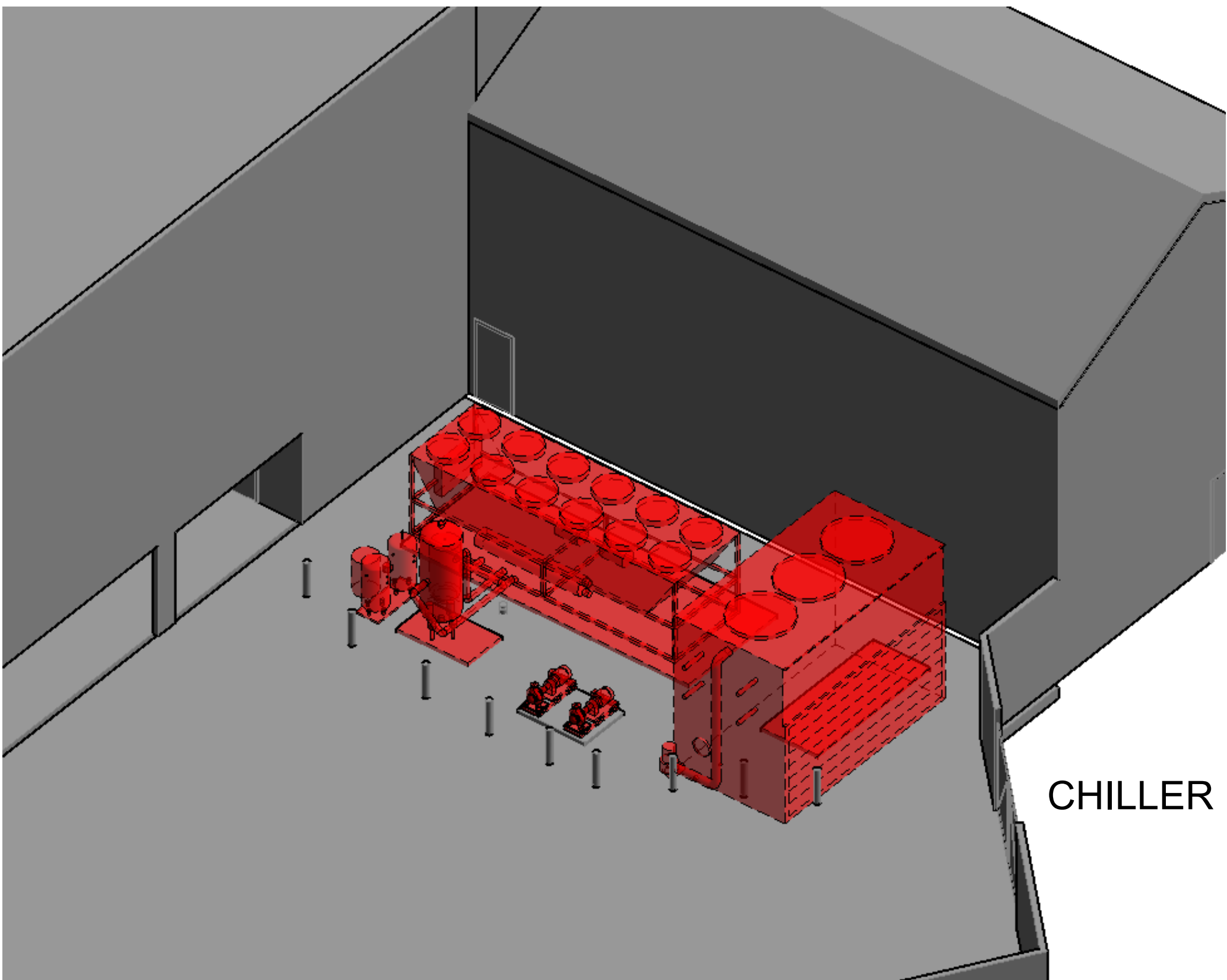
CHILLER REPLACEMENT AT USSL



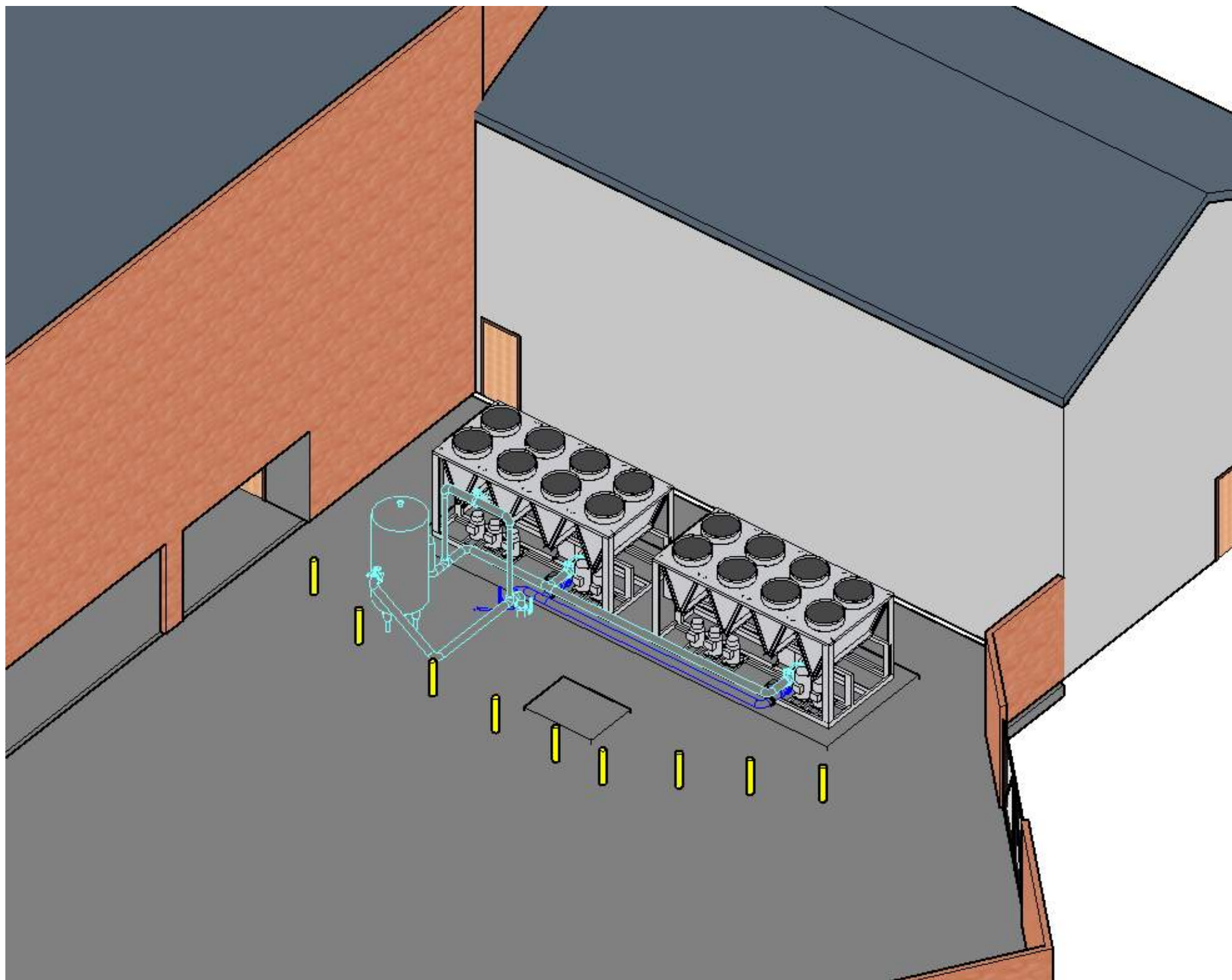
SITE LOCATION MAP



VICINITY MAP



CHILLER DEMOLITION



CHILLER REPLACEMENT AT USSL

SHEET LIST

SHEET COUNT	SHEET NUMBER	Sheet Name
00 GENERAL		
1 of 17	G-001	COVER SHEET
2 of 17	G-002	GENERAL NOTES, ABBREVIATIONS
01 ARCHITECTURAL		
3 of 17	A-100	LOWER LEVEL FLOOR PLAN
4 of 17	A-400	ENLARGED CHILLER PLANS
02 STRUCTURAL		
5 of 17	S-001	GENERAL NOTES AND TYPICAL DETAILS
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7 of 17	M001	MECHANICAL ABBREVIATION
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9 of 17	M100	MECHANICAL LOWER LEVEL OVERALL FLOOR PLAN
10 of 17	M101	MECHANICAL MID LEVEL OVERALL FLOOR PLAN
11 of 17	M201	MECHANICAL YARD FLOOR PLAN
12 of 17	M202	MECHANICAL ENLARGED PLAN
13 of 17	M300	MECHANICAL DETAIL
04 ELECTRICAL		
14 of 17	E001	GENERAL NOTES, LEGEND, AND ABBREVIATIONS
15 of 17	E100	ELECTRICAL DEMOLITION PARTIAL FLOOR PLAN
16 of 17	E101	ELECTRICAL PARTIAL POWER PLAN
17 of 17	E200	SINGLE LINE DIAGRAMS

SCOPE OF WORK

- REMOVE EXISTING FEEDER TO EXISTING CHILLER TO BE DEMOLISHED.
- REMOVE EXISTING DISCONNECT SWITCH SERVING EXISTING CHILLER, PUMPS, & COOLING TOWER TO BE DEMOLISHED.
- PROVIDE ELECTRICAL CONNECTION TO NEW CHILLERS AND PUMPS.
- PROVIDE NEW CIRCUIT BREAKERS IN EXISTING MAIN SWITCHBOARD TO SERVE NEW CHILLERS.
- PROVIDE COORDINATION STUDY TO NEW CIRCUIT BREAKERS.
- PROVIDE COORDINATION STUDY TO NEW CIRCUIT BREAKERS.
- EXTEND EXISTING 120V POWER CONNECTION TO NEW CHILLERS.



A-E FIRM

USDA

PROJ. MANAGER:

EPM

DRAWN BY:

PPM

CHECKED BY:

SAFETY & HEALTH

CONTRACT NO.:

REAL PROPERTY

MARK	DATE	BY	DESCRIPTION
REVISIONS			
USDA Agricultural Research Service			
US SALINITY LABORATORY			
UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA			
ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSL			
COVER SHEET			
PROJECT NO:	USDA AGRICULTURAL RESEARCH SERVICE	SHEET ID	G-001
Project Number	USDA/ARS, PA		
ISSUE DATE:	800 BUCHANNAN STREET		
2-26-2025	ALBANY, CA		
			SHT 1 OF 17

DRAWING ABBREVIATIONS

A LABEL	CLASS A DOOR	E LABEL	CLASS E DOOR	M	
A/C	AIR CONDITION	EA	EACH	MACH	MACHINE ROOM
A/C UNIT	AIR CONDITIONING UNIT	EF	EACH FACE	MATL	MATERIAL
AE	ARCHITECT/ENGINEER	EFS	EXTERIOR INSULATION AND FINISH SYSTEM	MAX	MAXIMUM
AB	ANCHOR BOLT	EJ	EXPANSION JOINT	MC	MOISTURE CONTENT
ACC	ACCESSIBLE	EL	EACH SIDE	MD	METAL DECK
ACS	AUTOMATIC CONTROL SYSTEM	EL	ELEVATION	MECH	MECHANICAL
ACS DR	ACCESS DOOR	ELEV	ELEVATOR	MECH RM	MECHANICAL ROOM
ACS PNL	ACCESS PANEL	ENTR	ENTRANCE	MEMB	MEMBRANE
ACT	ACOUSTICAL CEILING TILE	EPS	EXPANDED POLYSTYRENE BOARD (INSULATION)	MF	MILL FINISH
ADA	AMERICANS WITH DISABILITIES ACT	EQ	EQUAL	MFR	MANUFACTURER
ADMIN	ADMINISTRATION	EWC	ELECTRIC WATER COOLER	MID	MIDDLE
AF	ABOVE FINISHED FLOOR	EXP	EXPOSED	ML STD	MILITARY STANDARD
AFG	ABOVE FINISHED GRADE	EXT	EXTERIOR	MIN	MINIMUM, MINUTE
AFS	ABOVE FINISHED SLAB	EXT	EXTINGUISHER	MIRR	MIRROR
AGGR	AGGREGATE	EXT	EXTREXTERIOR GRADE	MISC	MISCELLANEOUS
AHU	AIR HANDLING UNIT	FA	FIRE ALARM	MLDG	MOLDING (MOLDING)
AIB	AIR INFILTRATION BARRIER	FAAP	FIRE ALARM ANNUNCIATOR PANEL	MO	MASONRY OPENING
ALT	ALTERNATE	FAS	BD FASCIA BOARD	MOD	MODIFY
ALUM	ALUMINUM	FB	BRFACE BRICK	ME	MOISTURE BARRIER
ANOD	ANODIZE	FCO	FLOOR CLEANOUT	MTG	MOUNTING
APC	ACOUSTICAL PANEL CEILING	FD	FLOOR DRAIN	MTL	METAL
APPROX	APPROXIMATE	FDN	FOUNDATION	MVBL	MOVABLE
AR	AS REQUIRED	FE	FIRE EXTINGUISHER	MWP	MEMBRANE WATERPROOFING
ARCH	ARCHITECT	FEC	FIRE EXTINGUISHER CABINET	N	
ASC	ABOVE SUSPENDED CEILING	FED	FEDERAL	N	NORTH
ASSY	ASSEMBLY	FF	FINISH FACE	NPA	NOT APPLICABLE
ATC	ARCHITECTURAL TILE CEILING	FF EL	FINISH FLOOR ELEVATION	NFA	NATIONAL FIRE PROTECTION ASSOCIATION
AVG	AVERAGE	FF INSUL	FOIL BACKED INSULATION	NIC	NOT IN CONTRACT
AW	ARCHITECTURAL WOODWORK	FGL	FIBERGLASS	NO	NUMBER
AWT	ARCHITECTURAL WALL TREATMENT	FH	FIRE HOSE	NOM	NOMINAL
B		FHP	FULL HEIGHT PARTITION	NP	NO PAINT
B LABEL	CLASS B DOOR	FIN	FINISH	NRC	NOISE REDUCTI
BALC	BALCONY	FIN BS	FINISH BOTH SIDES	O	
BB	BASEBOARD	FIN GR	FINISH GRADE	OA	OVERALL
BC	BOOKCASE	FIXT	FIXTURE	OC	ON CENTER
BD	BOARD	FLED	FOLDING	OD	OUTSIDE DIAMETER
BDRY	BOUNDARY	FLG	FLEXIBLE	OFD	OVERFLOW DRAIN
BFF	BELOW FINISH FLOOR	FLM	FLOORING	OFF	OFFICE
BHMA	BUILDER'S HARDWARE	FLMT	FLUSH MOUNT	OGL	OBSCURE GLASS
BM	MANUFACTURER'S ASSOCIATION	FLR	FLOOR	OPH	OPPOSITE HAND
BLDGD	BUILDING	FM	FACTORY MUTUAL	OPNG	OPENING
BLKG	BLOCKING	FOC	FACE OF CONCRETE	OPP	OPPOSITE
BLT IN	BUILT-IN	FOM	FACE OF MASONRY	OPQ	OPAQUE
BN	BULLNOSE	FR	FIRE RESISTANT	OWSJ	OPEN WEB STEEL JOIST
BOS	BOTTOM OF STEEL	FRG	FIBER REINFORCED GYPSUM FRMG	OPR	OPERABLE
BOT	BOTTOM	FRP	FIBERGLASS REINFORCED PLASTIC	ORD	OVERFLOW ROOF DRAIN
BP	BUILDING PAPER	FRTW	FIRE RETARDANT TREATED WOOD	ORIG	ORIGINAL
BRKT	BRACKET	FS	FEDERAL SPECIFICATION	P	
BSMT	BASEMENT	FSNTR	FESTENER	PA	PUBLIC ADDRESS
BTWN	BETWEEN	FT	FEET	PAR	PARAPET
BUR	BUILT-UP ROOFING	FTG	FOOTING	PAT	PATTERN
C		FWC	FABRIC WALLCOVERING	PB	PULL BOX
C CONC	CLASS C CONCRETE	G		PBC	PARTICLEBOARD
C LABEL	CLASS C DOOR	G	NATURAL GAS	PBD	PRECAST CONCRETE
CAB	CABINET	GALV	GALVANIZED	PCF	POUNDS PER CUBIC FOOT
CATW	CATWALK	GB	GRAB BAR	PCT	PERCENT
CAV	CAVITY	GFCI	GOVERNMENT FURNISHED	PERF	PERFORATED
CCB	CEMENTITIOUS (BACKER) BOARD	GFGI	GOVERNMENT INSTALLED	PERIM	PERIMETER
CD	CONSTRUCTION DOCUMENTS	GFGI	GOVERNMENT INSTALLED	PI	PIASE
CDW	CHILLED DRINKING WATER	GFRG	GLASS-FIBER-REINFORCED GYPSUM	PL	PLASTER
CEM PLAS	CEMENT PLASTER	GLZ	GLAZING	PL	PROPERTY LINE
CER	CERAMIC	GR FL	GROUND FLOOR	PL GL	PLATE GLASS
CF	CONTRACTOR FURNISHED	GUT	GUTTER	PLAM	PLASTIC LAMINATE
CF/CI	CONTRACTOR	GYP BD	GYPSUM BOARD	PLAS	PLASTER
FURNISHED/CONTRACTOR INSTALLED		GYP PLAS	GYPSUM PLASTER	PLBG	PLUMBING
FURNISHED/CONTRACTOR FURNISHED EQUIPMENT		H		PLNG	PILING
CFLG	COUNTERFLASHING	HB	HOSE BIBB	PLYWD	PLYWOOD
CFM	CUBIC FEET PER MINUTE	HDPE	HIGH DENSITY POLYETHYLENE	PNL	PANEL
CFMF	COLD-FORMED METAL FRAMING	HDW	HARDWARE	PP PL	PUSH/PULL PLATE
CFCS	CUBIC FEET PER SECOND	HDWD	HARDWOOD	PR	PAIR
CG	CORNER GUARD	HEPA	HIGH EFFICIENCY PARTICULATE AIR (FILTER)	PRCST	PRECAST
CI	CAST IRON	HM	HOLLOW METAL	PRKG	PARKING
CIP	CAST-IN-PLACE	HMD	HOLLOW METAL DOOR	PS CONC	PRESTRESSED CONCRETE
CJ	CONTROL JOINT	HORIZ	HORIZONTAL	PSF	POUNDS PER SQUARE FOOT
CL	CEILING LINE	HT	HEIGHT	PSI	POUNDS PER SQUARE INCH
CLG	CEILING	HYDR	HYDRAULIC	PT	PRESSURE TREATED
CLG DIFF	CEILING DIFFUSER	I		PTD	PAPER TOWEL DISPENSER
CLG HT	CEILING HEIGHT	IBC	INTERNATIONAL BUILDING CODE	PTDR	PAPER TOWEL DISPENSER AND

GENERAL NOTES

SB	SPLASH BLOCK
SCHED	SCHEDULE
SD	SMOKE DETECTOR
SF	SQUARE FOOT (FEET)
SFTWD	SOFTWOOD
SGL	SINGLE
SHT MTL FLASH	SHEET METAL (FLASHING)
SHTG	SHEATHING
SHV	SHELVING
SIM	SIMILAR
SJ	SCORED JOINT
SKLT	SKYLIGHT
SLNT	SEALANT
SMK	SMOKE
SMLS	SEAMLESS
SP EL	SPOT ELEVATION
SPEC	SPECIFICATION
SQ	SQUARE
SQ IN	SQUARE INCH
SQ YD	SQUARE YARD
SST	STAINLESS STEEL
ST	STAIRS
STD	STANDARD
STL JST	STEEL JOIST
STL RF DK	STEEL ROOF DECK
STOR	STORAGE
STR	STRINGERS
STRB/HRN	STROBE/HORN
SUB FL	SUBFLOOR
SV	SHEET VINYL
SW	SIDEWALK
T	
T	TREAD
T/S	TUB/SHOWER
TC	TERRA COTTA
TD	TRENCH DRAIN
TEL	TELEPHONE
TEMP	TEMPORARY
TER	TERRAZZO
TFF	TOP OF FINISH FLOOR
THK	THICKNESS
TK BD	TACKBOARD
TMPD GL	TEMPERED GLASS
TN	TRUE NORTH
TOF	TOP OF FOOTING
TOM	TOP OF MASONRY
TOP	TOP OF PARAPET
TOPO	TOPOGRAPHY
TOS	TOP OF SLAB
TRANS	TRANSOM
TRTD	TREATED
TV	TELEVISION

INDEX OF DRAWINGS	
NUMBERING SYSTEM: DISCIPLINE: C - CIVIL L - LANDSCAPE A - ARCHITECTURE M - MECHANICAL E - ELECTRICAL P - PLUMBING S - STRUCTURAL K - FOOD SERVICE DISCIPLINE MODIFIER SHEET NUMBER WITHIN SERIES REFER TO A/E/C CAD STANDARD FOR SHEET NUMBERING STANDARDS	
ANNOTATION CALLOUTS/DRAWING SYMBOLS INTERIOR ELEVATION SHEET NUMBER ELEVATION NUMBER EXTERIOR ELEVATION SHEET NUMBER SECTION SHEET NUMBER ENLARGEMENT, DETAIL SHEET NUMBER PLAN NORTH NORTH ARROW w/ TRUE NORTH INDICATION ROOM TAG ROOM 101 SPOT ELEVATION 100' LEVEL NAME LEVEL 01 100' SHEET KEYNOTE A	

GENERAL NOTES
DRAWINGS AND DOCUMENT NOTES

1. THE INTENT OF THESE DRAWINGS AND SPECIFICATIONS IS TO INCLUDE ALL LABOR, MATERIALS AND SERVICES NECESSARY FOR COMPLETION OF ALL WORK SHOWN, PRESCRIBED, OR REASONABLY APPLIED BUT NOT LIMITED TO THAT EXPLICITLY INDICATED IN THE CONTRACT DOCUMENTS.

A. THESE PLANS HAVE BEEN PREPARED USING VISUAL INSPECTION AND MEASUREMENTS FROM THE ACCESSIBLE AREAS OF THE PROJECT WORK AREA.

THE CONTRACTOR SHALL VERIFY ALL EXISTING CONDITIONS.

B. THE A/E TEAM CANNOT GUARANTEE THE EXACT CONDITION OF ALL EXISTING WORK.

3. THE GENERAL CONTRACTOR (GC) AND ALL SUBCONTRACTORS SHALL VERIFY ALL DIMENSIONS AND CONDITIONS ON THE JOB SITE PRIOR TO BEGINNING CONSTRUCTION AND SHALL REPORT ANY DISCREPANCIES OR UNDEFINIED CONDITIONS TO THE COR FOR RESOLUTION BEFORE ANY WORK IS BEGUN.

2. DO NOT SCALE DRAWINGS.

3. DIMENSIONS ON DRAWINGS ARE SHOWN TO THE CENTERLINE OF COLUMNS AND TO FACE OF STUD AT WALLS AND PARTITIONS UNLESS NOTED OTHERWISE.

4. ALL WORK SHALL CONFORM TO ALL APPLICABLE CODES, STANDARDS, ORDINANCES AND REGULATIONS AS ADOPTED BY THE GOVERNMENT.

5. DIMENSIONS, DETAIL, NOTES AND /OR SYMBOLS THAT APPLY TO ONE UNIT, APPLY TO ALL UNITS IN LIKE SITUATIONS UNLESS NOTED OTHERWISE.

6. UPON AWARD OF THE CONTRACT, THE CONTRACTOR SHALL PROVIDE A CONSTRUCTION SCHEDULE, WHICH SHALL BE UPDATED AS REQUIRED. THE CONTRACTOR SHALL COORDINATE AND SUPERVISE THE WORK OF ALL TRADES AND IS RESPONSIBLE FOR ADHERENCE TO ALL CONTRACT DOCUMENTS, TIMELY PERFORMANCE OF THE WORK AND MAINTAINING THE JOB SITE IN A SAFE AND CLEAN MANNER.

7. THE CONTRACTOR SHALL BE RESPONSIBLE FOR MEANS, METHODS, TECHNIQUES AND PROCEDURES EMPLOYED IN PERFORMANCE OF WORK IN, ON OR AROUND THE JOB SITE. THE CONTRACTOR SHALL COORDINATE AND VERIFY ALL WORK PERFORMED BY SUBCONTRACTORS, INCLUDING COORDINATION FOR CONSTRUCTION MATERIAL AND EQUIPMENT DELIVERY AND LAY-DOWN AREA.

8. ALL CONTRACTORS AND SUBCONTRACTORS PERFORMING WORK ON, OR RELATED TO THIS PROJECT SHALL CONDUCT THEIR OPERATIONS SO THAT ALL EMPLOYEES ARE PROVIDED A SAFE WORK PLACE AND THE PUBLIC ARE PROTECTED. THE WORKPLACE SHALL COMPLY WITH THE "OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF THE US DEPARTMENT OF LABOR AND WITH ANY AND ALL OTHER APPLICABLE FEDERAL AND/OR LOCAL SAFETY REGULATIONS. THE CONTRACTOR SHALL ACQUIRE AND MAINTAIN RESPONSIBILITY FOR JOB SITE SAFETY CONDITIONS DURING THE COURSE OF DEMOLITION AND CONSTRUCTION OF THIS PROJECT. THIS REQUIREMENT SHALL APPLY CONTINUOUSLY AND NOT BE LIMITED TO NORMAL WORKING HOURS. THE CONTRACTOR SHALL DEFEND, INDEMNIFY AND HOLD HARMLESS THE GOVERNMENT AND ARCHITECT FROM ANY AND ALL LIABILITY, REAL OR ALLEGED, IN CONNECTION WITH PERFORMANCE OF WORK ON THIS PROJECT.

9. A. WHEREVER PRACTICAL, THE WORK AREAS SHALL BE FENCED, BARRICADED OR BLOCKED OFF FROM THE PUBLIC OR OCCUPANTS TO PREVENT UNAUTHORIZED ENTRY. EXITS SHALL NOT BE BLOCKED TO PREVENT EVACUATION OF THE BUILDING OCCUPANTS.

DEMOLITION NOTES

1. SEE DEMOLITION PLANS FOR ADDITIONAL NOTES AND INFORMATION.

2. MATERIALS, EQUIPMENT OR CONSTRUCTION NOT NOTED IN THE CONSTRUCTION DOCUMENTS ARE NOT PART OF THIS NEW WORK. IF DISCOVERED DURING THE COURSE OF THE DEMOLITION, THE CONTRACTOR SHALL REPORT TO THE CONTRACTING OFFICER FOR FURTHER INSTRUCTION.

CONTAINING MATERIALS (ACM) ARE PRESENT IN THE PROJECT AREA. REFER TO EMC LABS REPORT DATED 12/09/2022 FOR IDENTIFICATION OF PAINTS A

ASBESTOS AND LEAD BASED PAINT NOTES

1. ASBESTOS CONTAINING MATERIALS (ACM) ARE PRESENT IN THE PROJECT AREA. REFER TO EMC LABS REPORT DATED 12/09/2022 FOR IDENTIFICATION OF PAINTS AND MATERIALS UTILIZED IN HANGAR B1210.

2. PRIOR TO DEMOLITION AND CONSTRUCTION ACTIVITIES WHICH WOULD DISTURB THE IDENTIFIED ACM, THE ABATEMENT CONTRACTOR MUST MAINTAIN CURRENT LICENSES AS REQUIRED BY EAFB GUIDELINES, THE STATE AND LOCAL JURISDICTIONS FOR THE REMOVAL, TRANSPORTATION AND DISPOSAL OF ACM OR ANY OTHER REGULATED MATERIALS AND ACTIVITIES.

3. THE CONTRACTOR SHALL PROVIDE AN ASBESTOS ABATEMENT PLAN FOR THE CONTRACTING OFFICER TO REVIEW AND APPROVAL. THE CONTRACTING OFFICER SHALL HAVE NOT LESS THAN 10 DAYS FOR REVIEW AND APPROVAL. NO DEMOLITION WORK MAY COMMENCE PRIOR TO APPROVAL OF THE PLAN.

4. THE CONTRACTOR SHALL ABATE AND DISPOSE OF ALL HAZARDOUS MATERIALS FOUND AND/OR CONSTRUCTION, INCLUDING BUT NOT LIMITED TO ACM, IN ACCORDANCE WITH ALL APPLICABLE REGULATIONS.

5. THE CONTRACTOR SHALL HIRE AN INDEPENDENT THIRD PARTY AIR TESTING / MONITOR DURING ACM ABATEMENT WORK TO VERIFY OTHER PORTIONS OF THE BUILDING ARE NOT SUBJECT TO CONTAMINATION. THE CONTRACTOR SHALL COORDINATE PERFORMANCE OF WORK WITH THIRD PARTY AIR TESTING / MONITORING THROUGHOUT THE PROJECT.

6. ACM ABATEMENT SCOPE IS LIMITED TO THE REMOVAL OF ALL EXISTING MATERIALS THAT ARE, OR WILL BECOME EXPOSED DUE TO DEMOLITION OR CONSTRUCTION ACTIVITIES AS SHOWN WITHIN THE PROJECT DOCUMENTS.

7. APPLICABLE LAWS AND REGULATIONS REGARDING THE PRESENCE OF ASBESTOS SHALL BE FOLLOWED, INCLUDING PROVISIONS REQUIRING NOTIFICATION TO REGULATORY AGENCIES, BUILDING OCCUPANTS, RENOVATION CONTRACTORS AND WORKERS.

8. AN IN-PLACE OPERATIONS AND MAINTENANCE PROGRAM SHALL BE PREPARED BY THE ABATEMENT CONTRACTOR FOR ANY REMAINING INTACT ACM FOLLOWING BUILDING REMODEL WORK.

9. LEAD BASED PAINT WILL LIKELY BE DISTURBED DURING DEMOLITION OR RENOVATION AND MUST BE REMOVED IF IT WILL NOT REMAIN INTACT AND IN GOOD CONDITION THROUGHOUT THE DEMOLITION PROCESS. THE SURVEY NOTED IN ITEM 1 ABOVE INCLUDES THE SURVEY FOR LEAD-BASED PAINT. AS AN ALTERNATIVE TO THE INFORMATION IN THE SURVEY, ALL PAINT MAY BE ASSUMED TO CONTAIN LEAD AND BE TREATED AS LEAD BASED PAINT. IF ANY LEAD BASED PAINT IS REMAINING AFTER DEMOLITION/RENOVATION, IT MUST BE INTACT AND IN GOOD CONDITION, AND A LEAD OPERATIONS AND MAINTENANCE (O&M) PLAN MUST BE MAINTAINED BY THE GOVERNMENT.

10. THE NATIONAL EMISSIONS STANDARDS FOR HAZARDOUS AIR POLLUTION (NESHAP), REQUIRES REGULATED ASBESTOS CONTAINING MATERIAL (RCM) BE IDENTIFIED AND REMOVED PRIOR TO DEMOLITION OR RENOVATION IF THE RCM IS FRIABLE OR WOULD BECOME FRIABLE DURING THE DEMOLITION.

11. ALL HAZARDOUS WASTE SHALL BE TRANSPORTED AND COLLECTED, TEMPORARILY STORED, PACKAGED AND LABELED PER GOVERNMENT GUIDELINES.

B. STORE, POSITION AND USE EQUIPMENT, TOOLS, MATERIALS, SCRAPS AND TRASH IN A MANNER THAT DOES NOT PRESENT A HAZARD TO THE PUBLIC OR OCCUPANTS BY ACCIDENTAL SHIFTING, IGNITION OR OTHER HAZARDOUS ACTIVITY.

C. STORE AND TRANSPORT REFUSE AND DEBRIS IN A MANNER TO PREVENT UNSAFE OR UNHEALTHY CONDITIONS FOR THE PUBLIC OR OCCUPANTS. COVER REFUSE CONTAINERS AND REMOVE REFUSE ON A FREQUENT BASIS AS ACCESSIBLE TO THE CONTRACTING OFFICER OR THEIR AUTHORIZED REPRESENTATIVE. USE TARPAPLANS OR OTHER MEANS TO PREVENT LOOSE MATERIALS FROM DROPPING OUT OF TRUCKS IN TRANSPORTATION.

11. ALL LEGAL EXITS SHALL BE MAINTAINED DURING CONSTRUCTION AND, IN THEIR FINAL CONSTRUCTION, ALL EXITS SHALL BE OPERABLE AND HAVE CODE COMPLIANT HARDWARE FROM INSIDE SPACES WITHOUT THE USE OF A KEY, SPECIAL EFFORT OR KNOWLEDGE. EXITS SHALL BE PROVIDED WITH ILLUMINATED "EXIT" SIGNS AS REQUIRED BY CODES AND STANDARDS APPLICABLE TO THIS PROJECT.

12. THE CONTRACTOR SHALL PROVIDE A WASTE MANAGEMENT PLAN TO COMPLY WITH THE FOLLOWING, OR MEET REQUIREMENTS OF THE AIFR FORCE, WHICHEVER IS MORE STRINGENT

RECYCLE OR SALVAGE 50% OF THE NON-HAZARDOUS CONSTRUCTION AND DEMOLITION WASTE BY DIVERTING MATERIAL FROM DISPOSAL THROUGH EFFICIENT USE OF RECYCLING, REUSE OR SALVAGE FOR FUTURE USE OR SALE.

D. DOCUMENTATION SHALL BE PROVIDED WHICH INCLUDES IDENTIFICATION OF THE WASTE MANAGEMENT AGENCY AND VERIFICATION THAT THE PERCENTAGE OF THE DEMOLITION AND CONSTRUCTION WASTE MATERIALS DIVERT FROM LANDFILL COMPLIES WITH THE XXXX EDITION OF CALGREEN, SECTION 5.408.1.

13. THE CONTRACTOR SHALL IMMEDIATELY NOTIFY THE GOVERNMENT SHOULD UNIDENTIFIED FIELD CONDITIONS BE DISCOVERED.

14. THE CONTRACTOR SHALL LIMIT THE NUMBER OF SYSTEMS SHUT DOWNS TO THE MINIMUM EXTENT POSSIBLE TO COMPLETE THE WORK. THE CONTRACTING OFFICER MUST BE NOTIFIED 72 HOURS PRIOR TO ANY SYSTEM SHUT-DOWNS.

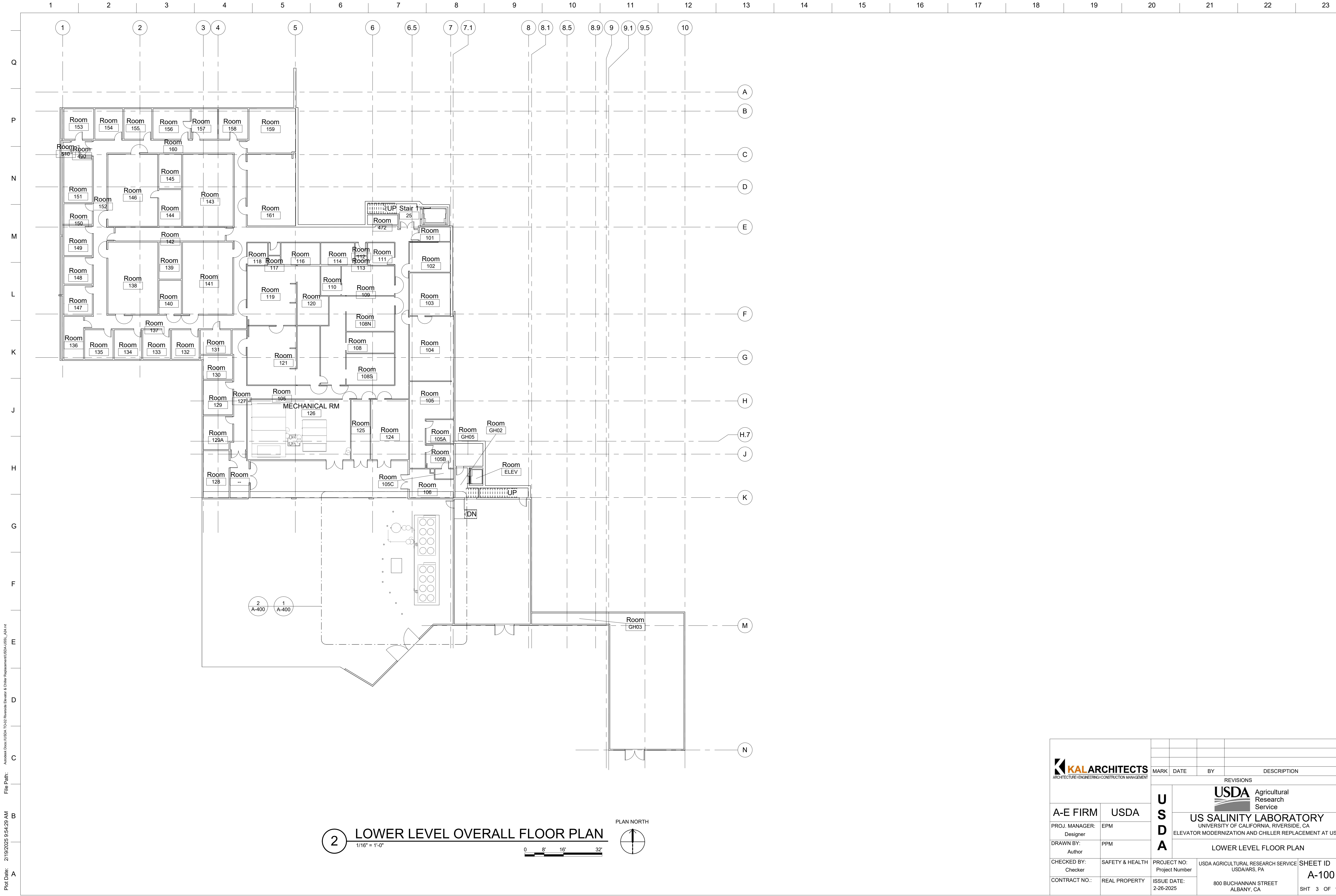
15. THE CONTRACTOR IS TO MAINTAIN THE INTEGRITY OF ALL BUILDING SYSTEMS (STRUCTURAL, MECHANICAL, PLUMBING, ELECTRICAL AND FIRE PROTECTION) THAT ARE UNCOVERED DURING THE DEMOLITION AND REMOVAL OF THE EXISTING BUILDING. IN THE DEMOLITION DOCUMENTS, THE CONTRACTOR SHALL PATCH OR REPAIR TO LIKE NEW CONDITION ANY BUILDING ELEMENTS DAMAGED AS A RESULT OF HIS WORK IN THE PROCESS OF CONSTRUCTION OR DEMOLITION AT NO ADDITIONAL COST TO THE GOVERNMENT.

16. THE CONTRACTOR SHALL BE RESPONSIBLE FOR LEAVING ALL FINISHED SURFACES CLEAN AT THE COMPLETION OF THE WORK, AND SHALL REMOVE ALL EXCESS MATERIAL AND DEBRIS FROM THE JOB SITE DURING CONSTRUCTION.

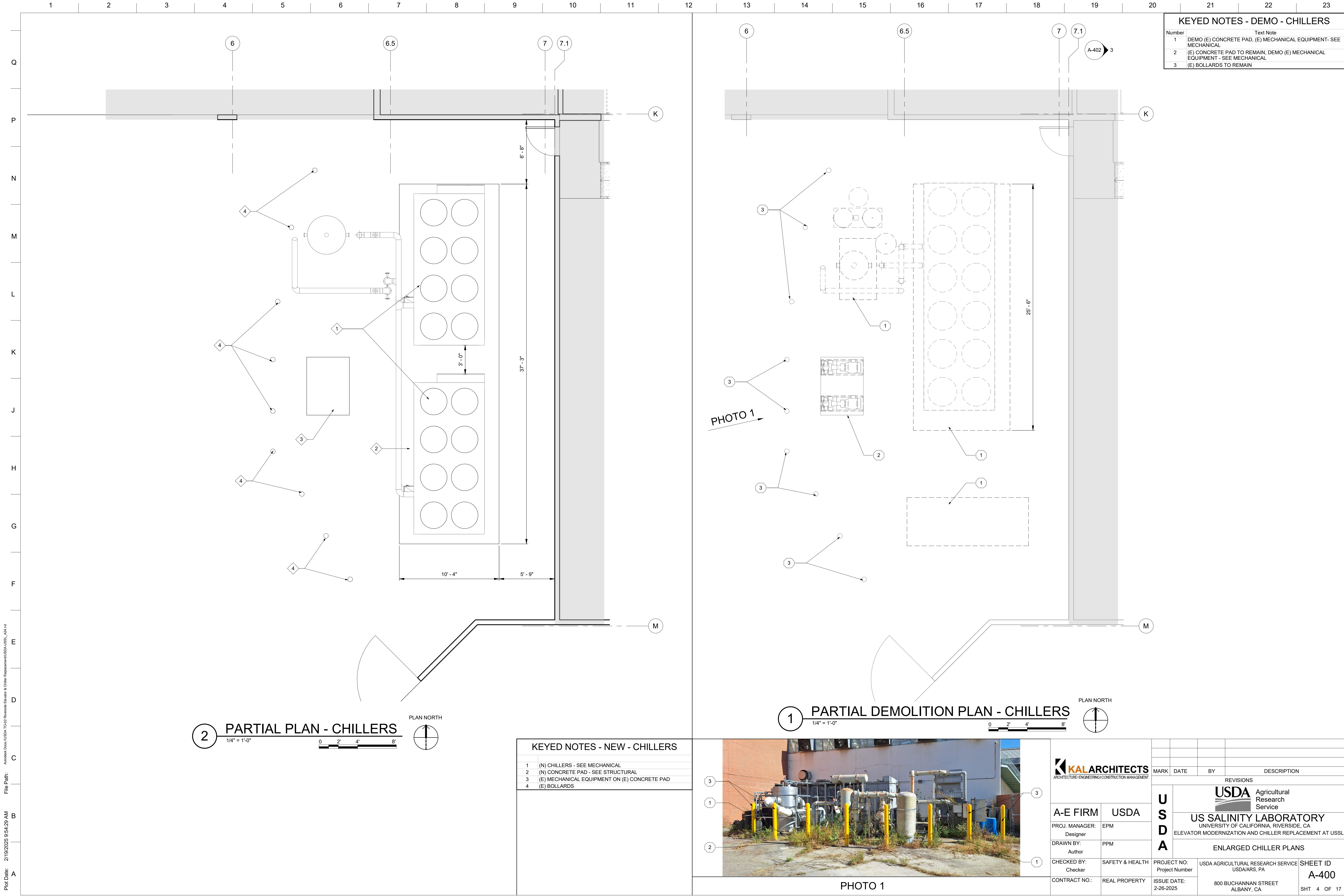
17. THE CONTRACTOR SHALL KEEP, ON SITE, A SET OF DRAWINGS AND SPECIFICATIONS MARKED AS AN UP-TO-DATE RECORD OF MODIFICATIONS OF THE CONSTRUCTION DOCUMENTS. UPON COMPLETION OF THE WORK, THE MARKED-UP SET OF "AS BUILT" CONDITIONS SHALL BE RETURNED TO THE GOVERNMENT COMPLETELY AND UNLESS OTHERWISE POSTED BY THE CONTRACTING OFFICER. CORRECTIONS AND REVISIONS IN THE ACTUAL CONSTRUCTION OF THE PROJECT, THESE RECORD DRAWINGS SHALL BE SIGNED BY SUBCONTRACTORS REPRESENTING ALL MAJOR TRADES FOR THE PROJECT AS WELL AS THE GENERAL CONTRACTOR.

18. GOVERNMENT REVIEW OR APPROVAL OF SHOP DRAWINGS SHALL NOT RELIEVE THE GENERAL CONTRACTOR FROM RESPONSIBILITY FOR DEVIATIONS FROM DRAWINGS OR SPECIFICATIONS UNLESS THE GC HAS CALLED THE DEVIATION TO THE GOVERNMENT'S ATTENTION (ON OR OFF SITE) AT THE TIME OF THE REVIEW. THE GC SHALL BE RELIEVED OF RESPONSIBILITY FOR ERRORS OF ANY SORT IN THE SHOP DRAWINGS.

 ARCHITECTURE • ENGINEERING • CONSTRUCTION MANAGEMENT					
		MARK	DATE	BY	DESCRIPTION
		REVISIONS			
A-E FIRM	USDA	U S D A	 Agricultural Research Service		
			US SALINITY LABORATORY UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSF		
			GENERAL NOTES, ABBREVIATIONS		
CHECKED BY: Checker	SAFETY & HEALTH	PROJECT NO: Project Number	USDA AGRICULTURAL RESEARCH SERVICE USDA/ARS, PA	SHEET ID G-002	
CONTRACT NO.:	REAL PROPERTY	ISSUE DATE: 2-26-2025	800 BUCHANNAN STREET ALBANY, CA	SHT 2 OF 17	



KALARCHITECTS ARCHITECTURE • ENGINEERING • CONSTRUCTION MANAGEMENT A-E FIRMUSDA PROJ. MANAGER: EPM Designer DRAWN BY: PPM Author CHECKED BY: SAFETY & HEALTH Checker CONTRACT NO.: REAL PROPERTY		USDA Agricultural Research Service US SALINITY LABORATORY UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSL	
		LOWER LEVEL FLOOR PLAN	
		PROJECT NO: USDA AGRICULTURAL RESEARCH SERVICE Project NumberUSDA/ARS, PA	SHEET ID A-100
		ISSUE DATE: 2-26-2025	800 BUCHANNAN STREET ALBANY, CA SHT 3 OF 17
MARK		DATE	BY
REVISIONS		DESCRIPTION	
U			
S			
D			
A			



KEYED NOTES - DEMO - CHILLERS	
Number	Text Note
1	DEMO (E) CONCRETE PAD, (E) MECHANICAL EQUIPMENT- SEE MECHANICAL
2	(E) CONCRETE PAD TO REMAIN, DEMO (E) MECHANICAL EQUIPMENT - SEE MECHANICAL
3	(E) BOLLARDS TO REMAIN

KEYED NOTES - NEW - CHILLERS	
1	(N) CHILLERS - SEE MECHANICAL
2	(N) CONCRETE PAD - SEE STRUCTURAL
3	(E) MECHANICAL EQUIPMENT ON (E) CONCRETE PAD
4	(E) BOLLARDS

1 PARTIAL DEMOLITION PLAN - CHILLERS
1/4" = 1'-0"

2 PARTIAL PLAN - CHILLERS
1/4" = 1'-0"



PHOTO 1



A-E FIRM	USDA
PROJ. MANAGER:	EPM
Designer	
DRAWN BY:	PPM
Author	
CHECKED BY:	SAFETY & HEALTH
Checker	
CONTRACT NO.:	REAL PROPERTY

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REVISIONS				
U S D A			Agricultural Research Service	
			US SALINITY LABORATORY	
	UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA			
	ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSL			
ENLARGED CHILLER PLANS				
PROJECT NO:		USDA AGRICULTURAL RESEARCH SERVICE		SHEET ID
Project Number		USDA/IARS, PA		A-400
ISSUE DATE:		800 BUCHANNAN STREET ALBANY, CA		SHT 4 OF 17
2-26-2025				

File Path: Plot Date:

GENERAL NOTES

- ALL MATERIALS AND WORKMANSHIP SHALL CONFORM TO PROJECT SPECIFICATIONS, PLANS AND ALL GOVERNING ORDINANCES.
- ALL WORK SHALL COMPLY WITH THE REQUIREMENT OF INTERNATIONAL BUILDING CODE 2021, AND ALL OTHER LOCAL OR STATE AGENCIES HAVING JURISDICTION OVER THIS PROJECT.
- CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND CONDITIONS OF THE JOB PRIOR TO STARTING CONSTRUCTION AND THE OWNER, ARCHITECT OR ENGINEER SHALL BE NOTIFIED IMMEDIATELY OF ANY DISCREPANCIES OR INCONSISTENCIES. DO NOT SCALE DIMENSIONS. WHERE NO DIMENSIONS ARE PROVIDED, CONSULT WITH OWNER, ARCHITECT OR ENGINEER FOR CLARIFICATION BEFORE PROCEEDING WITH THE WORK.
- ANY WORK PERFORMED IN CONFLICT WITH THE CONTRACT DOCUMENTS OR ANY CODE REQUIREMENTS SHALL BE CORRECTED BY THE CONTRACTOR AT HIS OWN EXPENSE AND AT NO EXPENSE TO THE OWNER, ARCHITECT OR ENGINEER.
- ALL SYMBOLS AND ABBREVIATIONS USED ON THE DRAWINGS ARE CONSIDERED TO BE CONSTRUCTION STANDARDS. IF CLARIFICATION IS REQUIRED, THE CONTRACTOR SHALL NOTIFY THE ARCHITECT OR ENGINEER PRIOR TO PROCEEDING WITH THE WORK.
- NO OPENINGS, POCKETS, NOTCHES, BLOCKOUTS, ETC., ARE ALLOWED IN STRUCTURAL ELEMENTS UNLESS DETAILED ON THE STRUCTURAL DRAWINGS. NOTIFY THE ARCHITECT OR ENGINEER OF ANY OPENINGS, POCKETS, ETC., SHOWN BY OTHER DRAWINGS BUT NOT SHOWN ON STRUCTURAL DRAWINGS.
- THE CONTRACTOR SHALL PROVIDE ALL MEASURES NECESSARY TO PROTECT THE STRUCTURE DURING CONSTRUCTION. SUCH MEASURES SHALL INCLUDE BUT ARE NOT LIMITED TO BRACING, SHORING, ETC. OBSERVATION VISITS TO THE SITE BY OWNER, ARCHITECT OR ENGINEER SHALL NOT INCLUDE THE INSPECTION OF BRACING, SHORING, ETC.
- CONSTRUCTION MATERIAL SHALL BE SPREAD OUT IF PLACED ON FRAMED FLOORS, ROOFS, ETC. THE LOAD SHALL NOT EXCEED THE DESIGN LIVE LOAD.
- THE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL EXCAVATION PROCEDURES INCLUDING LAGGING, SHORING, AND PROTECTION OF ADJACENT PROPERTY, STRUCTURES, STREET AND UTILITIES IN ACCORDANCE WITH ALL NATIONAL, STATE AND LOCAL SAFETY ORDINANCES.
- SEE MECHANICAL, ELECTRICAL AND/OR ARCHITECTURAL DRAWINGS FOR LOCATION AND SIZE OF PIPE, VENT, DUCT AND OTHER SIMILAR OPENINGS, AND EMBEDDED OR ATTACHED ITEMS.
- NOTES AND DETAILS ON DRAWINGS SHALL TAKE PRECEDENCE OVER GENERAL NOTES AND TYPICAL DETAILS.
- DO NOT SCALE DRAWINGS.

CONCRETE

- CONCRETE SHALL BE AS FOLLOWS:

LOCATION	AGGREGATES	STRENGTH @ 28 DAYS	MAX W/CM RATIO	DRY WEIGHT
FOOTINGS	HARDROCK	3000 PSI	0.50	145 PCF
SLAB-ON-GRADE	HARDROCK	4000 PSI	0.50	145 PCF
EQUIP. PADS	HARDROCK	4000 PSI	0.50	145 PCF

NOTE: A HIGHER GRADE OF CONCRETE MAY BE SUBSTITUTED FOR THOSE SHOWN ABOVE, BUT WILL BE SUBJECT TO THE CODE REQUIREMENTS OF THE HIGHER GRADE.

- CEMENT SHALL CONFORM TO ASTM C150 TYPE I OR II.
- HARDROCK AGGREGATES SHALL CONFORM TO ASTM C33. THEIR MAXIMUM SIZE SHALL BE 1-1/2 INCHES FOR FOOTINGS GRADE BEAMS AND 1 INCH FOR ALL OTHER WORK.
- LIGHTWEIGHT AGGREGATES (CONFORMING TO ASTM C330) SHALL BE APPROVED BY THE ENGINEER, AND THEIR MAXIMUM SIZE SHALL BE ONE INCH.
- A STATEMENT OF MIX DESIGN SHALL BE MADE FOR CONCRETE WITH DESIGN STRENGTH OVER 2500 PSI. COPIES OF THE MIX DESIGN SHALL BE SUBMITTED TO THE OWNER, ARCHITECT, ENGINEER, AND CONTRACTING OFFICER FOR APPROVAL BEFORE USE.
- ONLY ONE GRADE OF CONCRETE SHALL BE POURED ON THE JOB AT ONE TIME.
- CONTINUOUS INSPECTION IS REQUIRED FOR ALL CONCRETE WITH A DESIGN STRENGTHS OVER 2000 PSI.
- CONCRETE COVER OVER REINFORCING SHALL BE AS FOLLOWS:
 - CAST AGAINST AND PERMANENTLY EXPOSED TO EARTH 3"
 - EXPOSED TO EARTH OR WEATHER
 - CAST-IN-PLACE - NON STRESSED 2"
 - NOT EXPOSED TO EARTH OR WEATHER 2"
 - MAIN BARS IN BEAMS AND COLUMNS 2"
 - TIES AND STIRRUPS 1 1/2"
- ALL REINFORCING SHOWN CONTINUOUS SHALL BE LAPPED PER ACI 318 CLASS B. SPLICE OR 40 DIAMETERS (GRADE 60) AT SPLICES, AND SHALL BE MADE AWAY FROM POINTS OF MAXIMUM STRESS. MINIMUM LAP SHALL BE 2'-0" LONG.
- BEFORE CONCRETE IS POURED, CHECK WITH ALL TRADES TO INSURE PROPER PLACEMENT OF ALL OPENINGS, SLEEVES, CURBS, CONDUITS, BOLTS, INSERTS, ETC., RELATING TO WORK.
- ALL SLEEVES NOT SPECIFICALLY SHOWN ON THE DRAWINGS SHALL BE LOCATED BY THE TRADES INVOLVED AND SHALL BE APPROVED BY THE ENGINEER.
- DRY PACK CONCRETE SHALL BE ONE PART PORTLAND CEMENT AND ONE PART SAND WITH SUFFICIENT WATER TO ALLOW A SMALL AMOUNT OF PASTE TO COME TO THE SURFACE.
- CONCRETE GROUT SHALL BE NON SHRINKING WITH SUFFICIENT WATER TO ALLOW POURING. ULTIMATE COMPRESSIVE STRENGTH (F'C) AT 28 DAYS SHALL BE EQUAL TO 3000 PSI MINIMUM.
- A MIX THAT PRODUCES THE LOWEST SLUMP COMPATIBLE WITH PROPER PLACEMENT SHALL BE USED (4-1/2" MAXIMUM SLUMP UNLESS OTHERWISE APPROVED BY THE STRUCTURAL ENGINEER.)
- ALL CONCRETE IS DESIGNED BY ULTIMATE STRENGTH DESIGN METHOD.
- FORMS FOR CONCRETE STRUCTURAL MEMBERS (SLABS, BEAMS, ETC.) SHALL NOT BE STRIPPED UNTIL THE CONCRETE HAS REACHED ITS DESIGNED STRENGTH.

REINFORCING

- ALL REINFORCING SHALL CONFORM TO ASTM A-615 SPECIFICATIONS, 2-2628(B) GRADE 60 EXCEPT #3 BARS MAYBE GRADE 40. WELDED BARS SHALL BE ASTM A-706
- REINFORCING BARS SHALL BE SPLICED AND BENT IN STRICT ACCORDANCE WITH THE DRAWINGS AND DETAILS AND C.R.S.I. PUBLICATIONS. NO KINKS ALLOWED. ALL BARS SHALL BE CLEAN PRIOR TO CONCRETE PLACEMENT.
- PROVIDE DOWELS OF SAME SIZE AND NUMBER FROM ADJACENT POUR. BOTH VERTICALLY AND HORIZONTALLY TO MATCH TYPICAL REINFORCING SHOWN. LAPS TO BE IN ACCORDANCE WITH THE DRAWINGS AND DETAILS. DOWELS SHALL BE CLEANED AFTER POUR.
- USE LOW HYDROGEN ELECTRODES, GRADE E-70 OR E-90, FOR WELDING OF REINFORCING BARS.
- SHOP DRAWINGS FOR REINFORCING STEEL SHALL BE SUBMITTED TO THE STRUCTURAL ENGINEER FOR APPROVAL PRIOR TO FABRICATIONS.
- FIELD WELDING OR BENDING OF REINFORCING IS NOT PERMITTED EXCEPT AS INDICATED ON THE DRAWINGS OR AS APPROVED BY THE STRUCTURAL ENGINEER.

DESIGN CRITERIA

- SEISMIC DESIGN DATA:

RISK CATEGORY II
SEISMIC IMPORTANCE FACTOR $I_e = 1.0$
MAPPED SPECTRAL RESPONSE: $S_S = 1.91$ $S_1 = 0.69$
SITE CLASSIFICATION: D
SPECTRAL RESPONSE: $S_{DS} = 1.35$
SEISMIC DESIGN CATEGORY: D

- WIND LOADS:

RISK CATEGORY II
BASIC WIND SPEED (3-SEC. GUST) = 96 MPH
WIND EXPOSURE = C

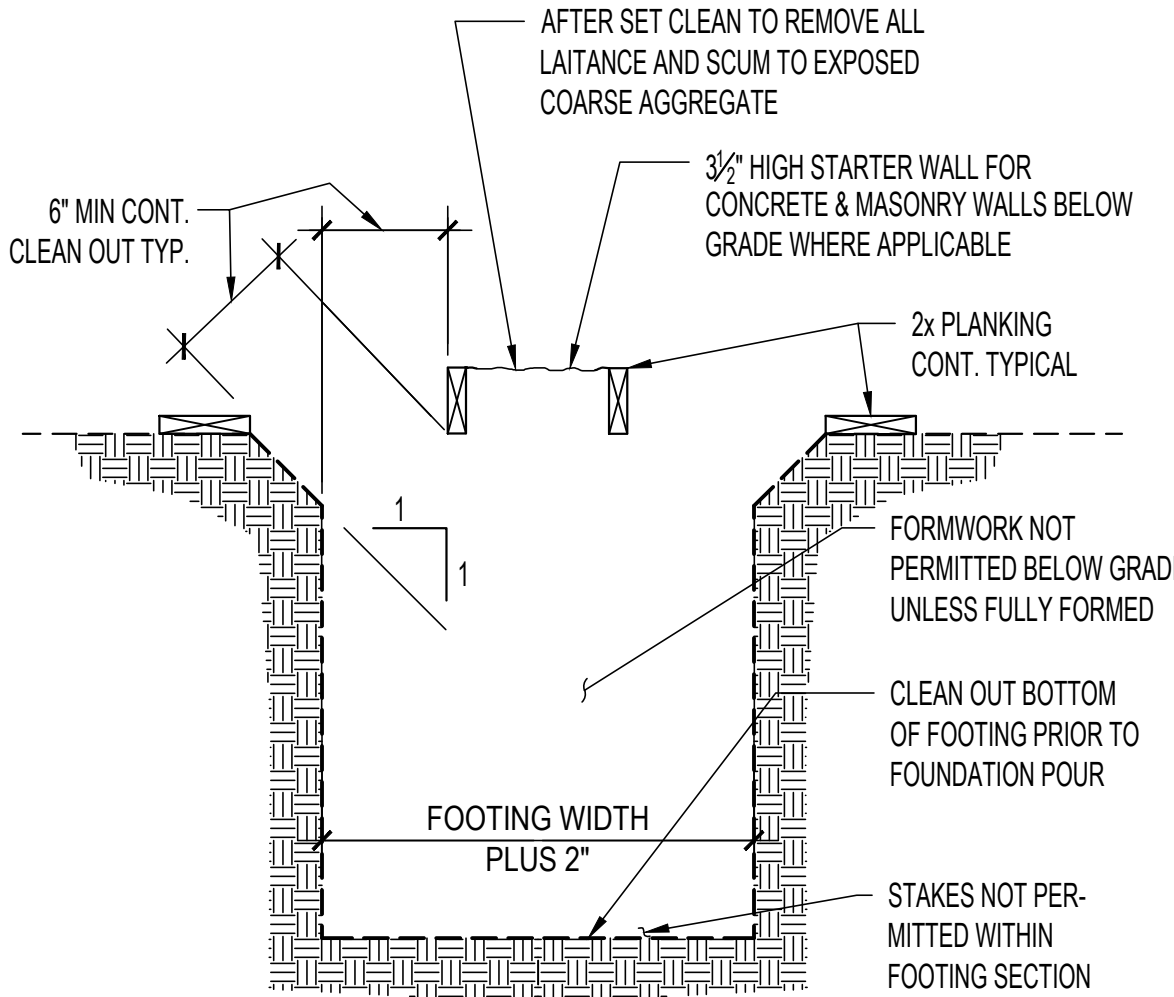
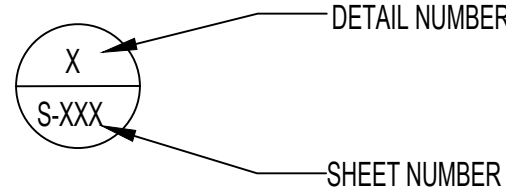
SPECIAL INSPECTION PER 2021 IBC

- SPECIAL INSPECTION PER IBC 2021 CHAPTER 17 BY A PERSON WHOSE QUALIFICATIONS ARE ACCEPTABLE TO THE STRUCTURAL ENGINEER OF RECORD AND SHALL BE REQUIRED FOR THE FOLLOWING WORK:
 - CONCRETE STRENGTH GREATER THAN 2500 psi
 - POST INSTALLED ANCHORS WEDGE AND EPOXY ANCHORS.
 - HIGH STRENGTH BOLTS
 - SHOP AND FIELD WELDING
 - CONCRETE MASONRY CONSTRUCTION
- INSPECTION REPORTS SHALL BE PROVIDED TO THE CONTRACTING OFFICER, ENGINEER, OR ARCHITECT OF RECORD WITHIN ONE WEEK OF INSPECTION
- THE SPECIAL INSPECTOR SHALL BE A QUALIFIED PERSON WHO DEMONSTRATES COMPETENCE TO THE SATISFACTION OF THE REGISTERED DESIGN PROFESSIONAL RESPONSIBLE FOR THE DESIGN OF THE THE STRUCTURE FOR INSPECTION OF THE PARTICULAR TYPE OF CONSTRUCTION OR OPERATION REQUIRING SPECIAL INSPECTION.
- DUTIES AND RESPONSIBILITIES OF THE SPECIAL INSPECTOR:
THE SPECIAL INSPECTOR SHALL OBSERVE THE WORK ASSIGNED FOR CONFORMANCE WITH:
 - THE APPROVED DESIGN DRAWINGS AND SPECIFICATIONS. THE INSPECTOR SHALL NOT ALTER, MODIFY, ENLARGE, OR WAIVE ANY OF THE REQUIREMENTS OF THE DOCUMENTS.
 - THE SPECIAL INSPECTOR SHALL FURNISH INSPECTION REPORTS TO THE CONTRACTING OFFICER, THE ROFESSIONAL-OF-RECORD, AND THE CONTRACTOR. ALL DISCREPANCIES SHALL BE BROUGHT TO THE IMMEDIATE ATTENTION OF THE CONTRACTOR FOR CORRECTION, THEN, IF UNCORRECTED, SUBMIT A COMPLETE LIST OF ALL OUTSTANDING DISCREPANCIES ON A WEEKLY BASIS TO THE CONTRACTING OFFICER, AND THE PROFESSIONAL-OF-RECORD, UNTIL ALL CORRECTIONS HAVE BEEN COMPLETED.
 - THE SPECIAL INSPECTOR SHALL SUBMIT A FINAL SIGNED REPORT STATING WHETHER THE WORK REQUIRING SPECIAL INSPECTION WAS, TO THE BEST OF THE INSPECTOR'S KNOWLEDGE, IN CONFORMANCE WITH THE APPROVED PLANS AND SPECIFICATIONS AND THE APPLICABLE WORKMANSHIP PROVISIONS OF THE BUILDING CODE.
- WHERE SPECIAL INSPECTION REQUIREMENTS DUPLICATE THE REQUIREMENTS OF OTHER SPECIFIED TESTING, DUPLICATE INSPECTIONS SHALL NOT BE REQUIRED

LEGEND

STD STANDARD
EA EACH
(N) NEW
TYP TYPICAL CONDITION
CONC CONCRETE
CLR. CLEAR
& AND
@ AT
W/ WITH
EQUIV EQUIVALENT
Ø DIAMETER
PL PLATE
AB ANCHOR BOLT

DETAIL REFERENCE



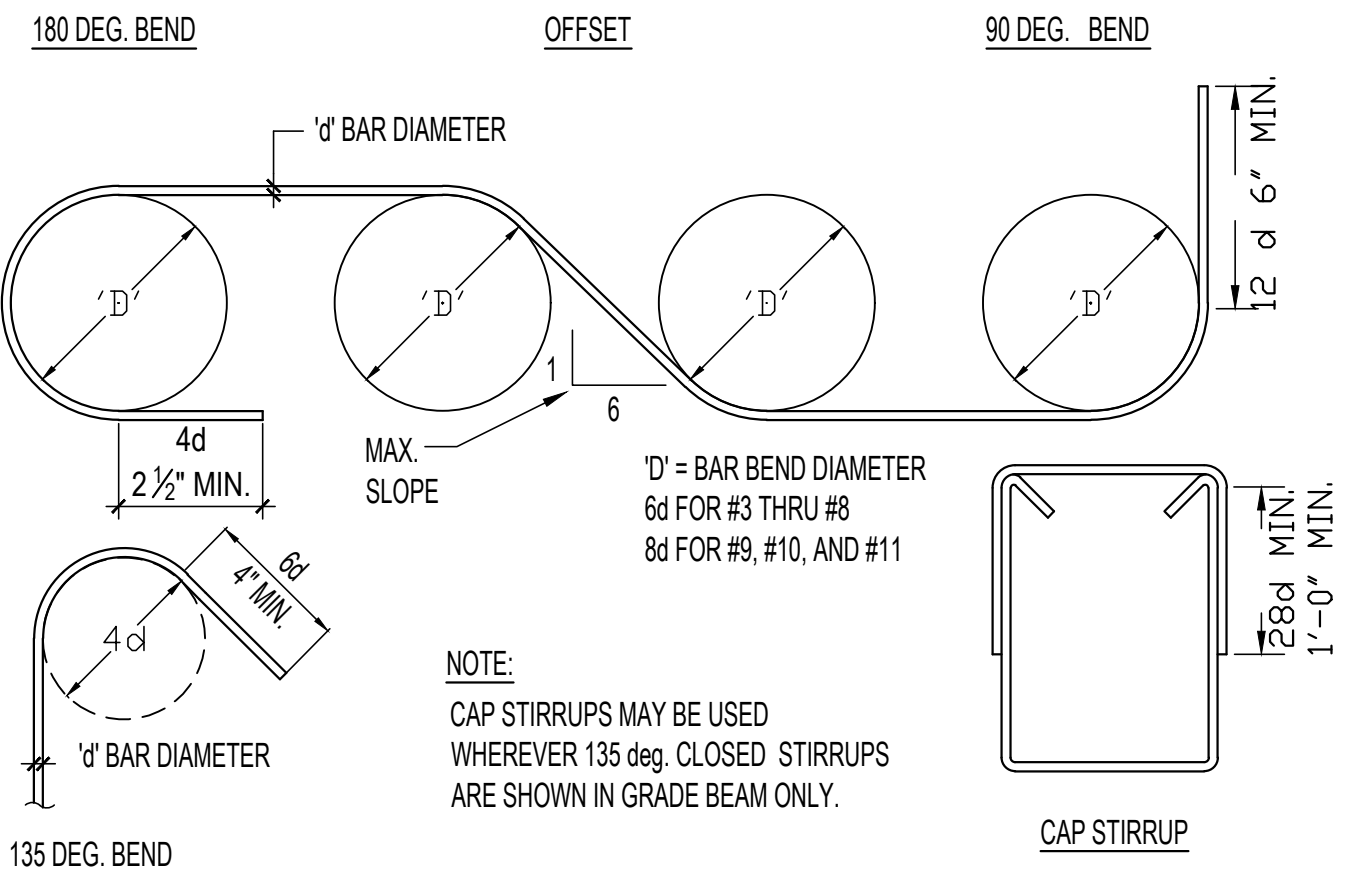
NOTES:

- FOUNDATION CONCRETE MAY BE PLACED DIRECTLY INTO EXCAVATION (AS SHOWN) PROVIDED THE TRENCH WALLS ARE STABLE AS DETERMINED BY THE SOIL ENGINEER.
- FOUNDATIONS SHALL BEAR ON ACCEPTABLE SOIL, PER THE SOIL REPORT, AS DETERMINED BY THE SOIL ENGINEER.

TYPICAL FOOTING TRENCH DETAIL

SCALE 1"=1'-0"

1



BAR DEVELOPMENT LENGTH (Ld) SCHEDULE				
BAR SIZE	F _c = 3000 PSI		F _c = 4000 PSI	
	TOP BARS	OTHER BARS	TOP BARS	OTHER BARS
#3	21	16	19	16
#4	29	22	25	19
#5	36	27	31	24
#6	43	33	37	29
#7	62	48	54	42
#8	70	55	62	48
#9	86	66	69	54
#10	105	81	77	60
#11	125	96	85	65

ACI CLASS A TENSION SPLICE = 1.0 L_d
LAP SPLICES FOR ACI CLASS B TENSION SPLICE = 1.3 L_d

TYPICAL REINFORCING BENDING & SPLICE DETAIL

SCALE 1"=1'-0"

5



A-E FIRM

USDA

PROJ. MANAGER:

EPM

Designer

DRAWN BY:

PPM

CHECKED BY:

SAFETY & HEALTH

CONTRACT NO.:

REAL PROPERTY

PROJECT NO:

Project Number

ISSUE DATE:

2-26-2025

USDA AGRICULTURAL RESEARCH SERVICE

USDA/ARS, PA

800 BUCHANAN STREET

ALBANY, CA

GENERAL NOTES AND

TYPICAL DETAILS

SHEET ID

S-001

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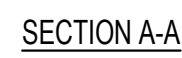
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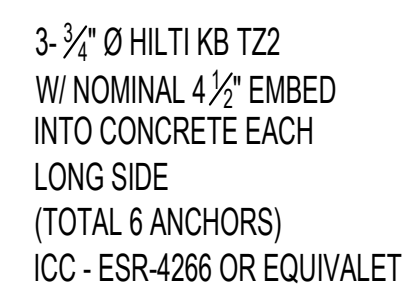
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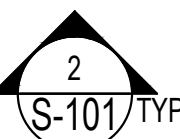
BUFFER TANK PAD AND ANCHORAGE DETAIL

SCALE	4
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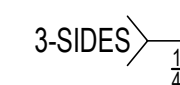
CHILLER 1 & 2 ANCHOR AND PAD SECTION

SCALE	
NTS	



PLAN CHILLER CONCRETE PAD

SCALE 1/4"=1'-0"	1
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BUFFER TANK ANGLE CONNECTION DETAIL

SCALE	
NTS	

 KALARCHITECTS ARCHITECTURE • ENGINEERING • CONSTRUCTION MANAGEMENT							
		MARK	DATE	BY	DESCRIPTION		
		REVISIONS					
A-E FIRM	USDA	U S D A			Agricural Research Service		
PROJ. MANAGER: EPM Designer			US SALINITY LABORATORY UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA				
DRAWN BY: PPM Author			ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSL				
CHECKED BY: SAFETY & HEALTH Checker			CHILLER FOUNDATION PLAN AND SECTION				
CONTRACT NO.:	REAL PROPERTY	PROJECT NO: Project Number	USDA AGRICULTURAL RESEARCH SERVICE USDA/ARS, PA			SHEET ID S-101	
		ISSUE DATE: 2-26-2025	800 BUCHANAN STREET ALBANY, CA			SHT 6 OF 17	

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STANDARD MECHANICAL SYMBOLS

PIPING LINETYPES LEGEND

	CONDENSATE DRAIN
	CONDENSATE RETURN
	CHILLED WATER RETURN
	CHILLED WATER SUPPLY
	HYDRONIC HOT WATER RETURN
	HYDRONIC HOT WATER SUPPLY
	REFRIGERANT LIQUID
	REFRIGERANT SUCTION
	STEAM
	PIPING WITH ABBREVIATION (ABV=TYPE; SEE ABBREVIATIONS)
	PIPING WITH ABBREVIATION UNDERGROUND (DOUBLE LINE)

PIPING SYMBOLS

SINGLE LINE	DOUBLE LINE	
		45 DEGREE DROP
		CONCENTRIC REDUCER
		FLEXIBLE CONNECTION
		PIPE CAP
		PIPE ELBOW DOWN
		PIPE ELBOW UP
		PIPE TEE DOWN
		PIPE TEE UP
		PITCH OF PIPE (R = RISE; D = DROP) IN DIRECTION OF FLOW

GENERAL SYMBOLS

	DRAWING NOTE HEX - NEW WORK
	DRAWING NOTE BOX - DEMOLITION
	CONNECT TO EXISTING / POINT OF WORK
	CALLOUT OUTLINE
	DETAIL OR SECTION BUBBLE
	DETAIL OR SECTION NUMBER
	SHEET NUMBER WHERE SECTION / DETAIL IS SHOWN
	EQUIPMENT MARK AND NUMBER

DUCTWORK LINETYPES LEGEND

	EXHAUST AIR
	FLUE VENT
	OUTSIDE AIR
	RETURN AIR
	SUPPLY AIR
	DUCTWORK WITH ABBREVIATION (ABV=TYPE; SEE ABBREVIATIONS)

PIPING ACCESSORIES SYMBOLS

SINGLE LINE	DOUBLE LINE	
		ABBREVIATION OF ITEM
		AUTOMATIC AIR VENT
		BALANCING VALVE
		BALL VALVE
		BUTTERFLY VALVE
		CALIBRATED BALANCING VALVE
		CHECK VALVE
		CHECK VALVE W/ BALL DRIP
		GATE VALVE
		GATE ANGLE VALVE
		GLOBE VALVE
		GLOBE ANGLE VALVE
		MANUAL AIR VENT
		METER
		PLUG VALVE
		PRESSURE GAGE & COCK
		PRESSURE REGULATING VALVE
		PRESSURE / TEMPERATURE PORT
		PUMP SUCTION DIFFUSER
		RELIEF VALVE
		STRAINER
		STRAINER (BLOW-OFF TYPE)
		THERMOMETER
		THERMOMETER WELL
		TRAP, STEAM W/ TYPE (F & T = FLOAT & THERMOSTAT; BT = BUCKET, TT = THERMOSTATIC)
		UNION

HVAC CONTROL SYMBOLS

	ANTI-TERRORISM FORCE PROTECTION SHUT-DOWN BUTTON
	CARBON MONOXIDE SENSOR
	CARBON DIOXIDE SENSOR
	COMBINATION HUMIDISTAT & THERMOSTAT
	HUMIDISTAT
	ROOM THERMOSTAT

NOTES:

- A. THIS LEGEND IS COMPOSED OF STANDARD SYMBOLS AND APPLIES TO THE SET OF DRAWINGS TO THE EXTENT APPLICABLE.
- B. SYMBOLS ARE REPRESENTATIONAL AND MAY NOT REFLECT ACTUAL CONDITIONS.
- C. REFER TO HVAC CONTROL SCHEMATIC DRAWINGS FOR ADDITIONAL CONTROL SYMBOLS & ABBREVIATIONS.

DUCTWORK ACCESSORIES SYMBOLS

SINGLE LINE	DOUBLE LINE	
		ACCESS DOOR - HORIZONTAL
		ACCESS DOOR - VERTICAL
		AUTOMATIC AIR DAMPER
		BACK DRAFT DAMPER
		COMBINATION FIRE/SMOKE DAMPER - VERTICAL POSITION (HORIZONTAL AIR FLOW)
		COMBINATION FIRE/SMOKE DAMPER - HORIZONTAL POSITION (VERTICAL AIR FLOW)
		FIRE DAMPER - VERTICAL POSITION (HORIZONTAL AIR FLOW)
		FIRE DAMPER - HORIZONTAL POSITION (VERTICAL AIR FLOW)
		MANUAL BALANCING DAMPER
		SMOKE DAMPER - VERTICAL POSITION (HORIZONTAL AIR FLOW)
		SMOKE DAMPER - HORIZONTAL POSITION (VERTICAL AIR FLOW)

DUCTWORK SYMBOLS

SINGLE LINE	DOUBLE LINE	
		EXHAUST W/ FLOW RATE (EG = GRILLE; ER = REGISTER) (200 = CFM QUANTITY)
		RETURN W/ FLOW RATE (RG = GRILLE; RR = REGISTER) (200 = CFM QUANTITY)
		SUPPLY AIR W/ FLOW RATE (SG = GRILLE; CD = CEILING DIFFUSER; SR = REGISTER) (200 = CFM QUANTITY)
		SUPPLY AIR W/ FLOW RATE AND W/ THROW DIRECTION (SG = GRILLE; CD = CEILING DIFFUSER; SR = REGISTER) (200 = CFM QUANTITY)
		SUPPLY AIR W/ FLOW RATE (LD = LINEAR DIFFUSER) (200 = CFM QUANTITY)
		SUPPLY AIR W/ FLOW RATE AND W/ THROW DIRECTION (LD = LINEAR DIFFUSER) (200 = CFM QUANTITY)
		EXHAUST AIR DUCT ELBOW DOWN AND UP
		RETURN AIR DUCT ELBOW DOWN AND UP
		SUPPLY AND OUTSIDE AIR DUCT ELBOW DOWN AND UP
		EXHAUST OR RETURN AIR FLOW DIRECTION
		SUPPLY AIR FLOW DIRECTION
		DUCT BRANCH TAKE-OFF - RECTANGULAR
		DUCT BRANCH TAKE-OFF - ROUND
		DUCT CAP
		DUCT PITCH IN DIRECTION OF FLOW (R = RISE; D = DROP)
		DUCT PRESSURE CLASS INDICATOR (INCHES W.G.)
		DUCT SIZE - RECTANGULAR; FIRST NUMBER SHOWN IS DIMENSION OF SIDE SHOWN (DIMENSIONS ARE CLEAR INTERNAL)
		DUCT SIZE - ROUND; NUMBER SHOWN IS DIAMETER DIMENSION OF DUCT (DIMENSIONS ARE CLEAR INTERNAL)
		FLEXIBLE DUCT CONNECTION

 ARCHITECTURE/ENGINEERING/CONSTRUCTION MANAGEMENT					
		MARK	DATE	BY	DESCRIPTION
		REVISIONS			
A-E FIRM		U S D A	 Agricultural Research Service		
USDA			US SALINITY LABORATORY		
PROJ. MANAGER: EPM Designer			UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSL		
DRAWN BY: PPM Author		MECHANICAL SYMBOLS & ABBREVIATION			
CHECKED BY: SAFETY & HEALTH Checker		PROJECT NO: Project Number	USDA AGRICULTURAL RESEARCH SERVICE USDA/IARS, PA		SHEET ID M001
CONTRACT NO.: REAL PROPERTY		ISSUE DATE: 2-26-2025	800 BUCHANNAN STREET ALBANY, CA		SHT 7 OF 17

Plot Date: 2/19/2025 1:17:54 PM File Path: Autocad Docs\USDA\1D-02\Riverside Elevator & Chiller Replacement\USDA\USSL_M002.rvt

AIR COOLED CHILLER SCHEDULE																																
MARK	LOCATION	AREA AND/OR BLDG SERVED	TYPE	CAPACITY		# OF COMP	COOLING EFFICIENCY (EER)	NPLV/IP	MAX IPLV (kW/TON)	EVAPORATOR								CONDENSER		ELECTRICAL										MANUFACTURER	MODEL NO	
										FLOW		EWT		LWT		MAX WPD		FOULING FACTOR	AMBIENT OA TEMP		COMPRESSOR MOTOR						TOTAL					
				TONS	[kW]					GPM	[L/s]	°F	[°C]	°F	[°C]	FT	[kPa]		°F	[°C]	# COMP	HP	[kW]	PHASE	VOLT	# FANS	MCA	MOCp	PHASE			VOLT
																											AMPS	AMPS				
CH-1&2	MECHANICAL YARD	WHOLE BUILDING	SCROLL	132		6	7.26	15.6	17.51	264		53		43		8.43		0.0001	95		6	35		3	460	8	261.4	300	3	460	CARRIER	30RC-1326S015-7-2

BUFFER TANK SCHEDULE																										
MARK	LOCATION	SYSTEM AND/OR SERVICE	TYPE	APPROX SYSTEM VOLUME		SYSTEM TEMPERATURE RANGE				INITIAL PRESSURE IN TANK		MAX OPERATING PRESSURE		FILL PRESSURE AT TANK				MIN VOLUME TANK		MIN BLADDER VOLUME		PIPE SIZE TO TANK		VENT SIZE		MANUFACTURER/MODEL NO
						MIN		MAX						RELIEF VALVE		AT TANK										
				GAL	[L]	°F	[°C]	°F	[°C]	PSIG	[kPa]	PSIG	[kPa]	PSIG	[kPa]	PSIG	[kPa]	GAL	[L]	GAL	[L]	IN	[mm]	IN	[mm]	
BT-1	MECHANICAL YARD	CHILLED WATER	VERT DIAPHRAGM	700		40		60		43		125		80		70		NA		NA		6		3/4"		WESSELS/CBT-700

PUMP SCHEDULE																						
MARK	LOCATION	AREA AND/OR BLDG SERVED	SYSTEM AND/OR SERVICE	TYPE	CIRCULATING FLUID										MIN % EFF	ELECTRICAL MOTOR					MANUFACTURER/MODEL NO	
					FLUID	FLOW		HEAD		NPSH AVAILABLE		TEMPERATURE		SP GR		NOMINAL POWER		PHASE	VOLT	MAX RPM		SPEED CONTROL
						GPM	[L/s]	FT	[kPa]	FT	[kPa]	°F	[°C]			HP	[kW]					
CHWP1&2	MECHANICAL ROOM 126	USDA Salinity Laboratory	CHILLED WATER	END-SUCTION	CHILLED WATER	530		100		9.9		55		92	80	20		3	460	1750	VARIABLE	TACO/F1011D



ARCHITECTURE|ENGINEERING|CONSTRUCTION MANAGEMENT

A-E FIRM

PROJ. MANAGER: Designer

DRAWN BY: Author

CHECKED BY: Checker

CONTRACT NO.:

USDA

EPM

PPM

SAFETY & HEALTH

REAL PROPERTY

MARK

DATE

BY

DESCRIPTION

REVISIONS

USDA Agricultural Research Service

US SALINITY LABORATORY

UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA

ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSL

MECHANICAL SCHEDULES

PROJECT NO:

ISSUE DATE:

USDA AGRICULTURAL RESEARCH SERVICE

USDA/IARS, PA

800 BUCHANNAN STREET

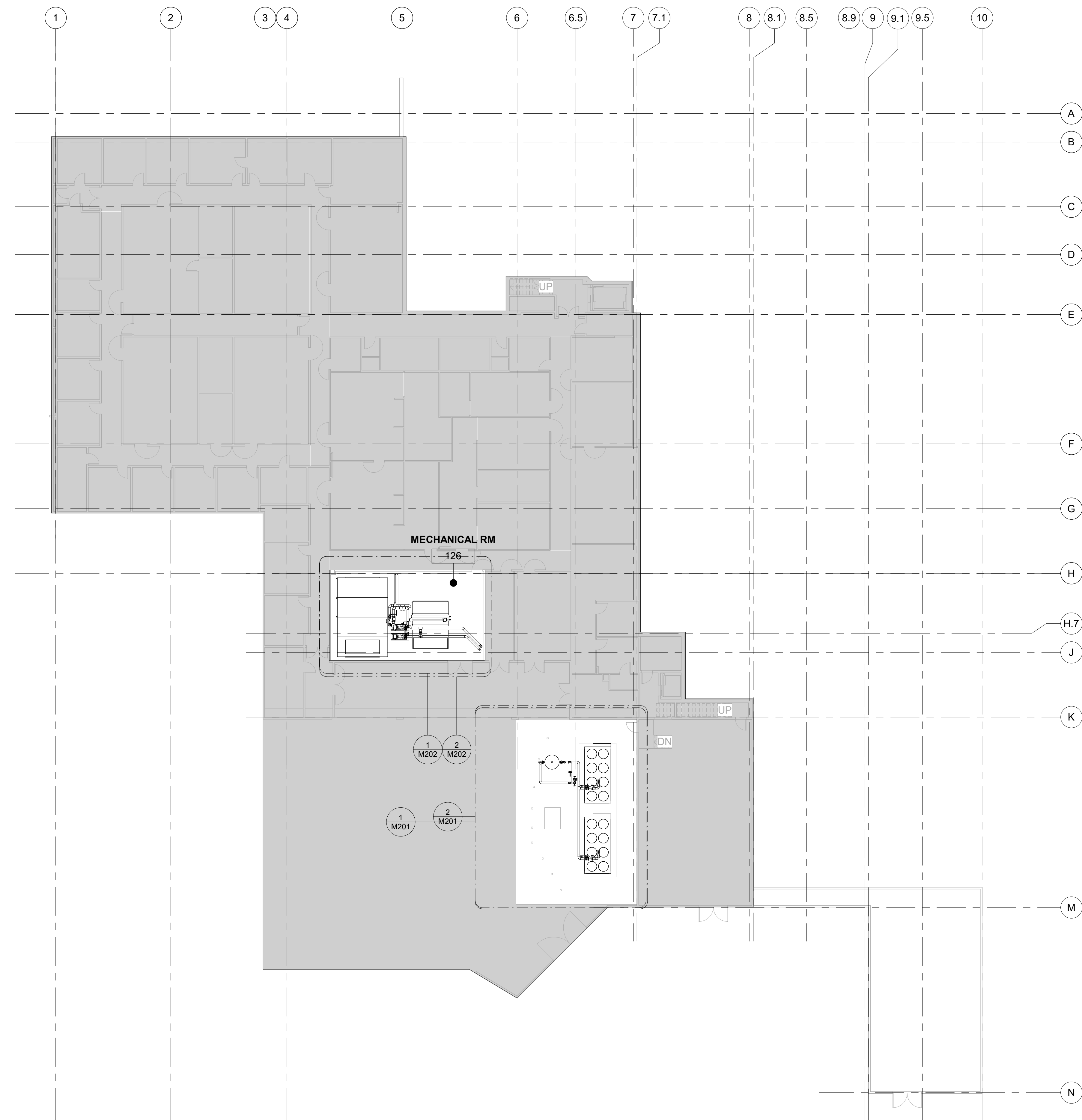
ALBANY, CA

SHEET ID

M002

SHT 8 OF 17

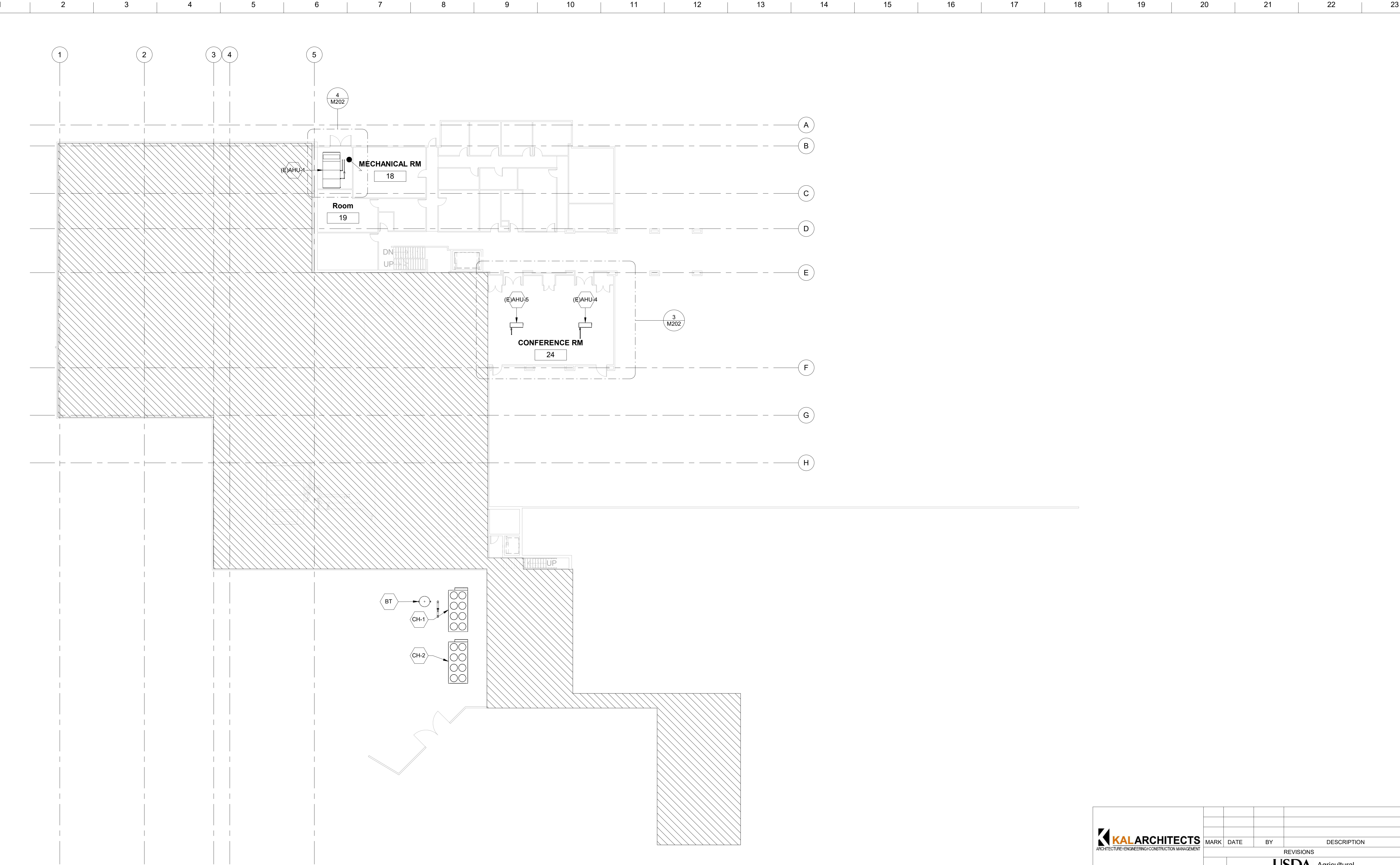
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1 MECHANICAL LOWER LEVEL OVERALL FLOOR PLAN
1/16" = 1'-0"

 KALARCHITECTS ARCHITECTUREENGINEERINGCONSTRUCTION MANAGEMENT A-E FIRM USDA PROJ. MANAGER: Designer EPM DRAWN BY: Author PPM CHECKED BY: Checker SAFETY & HEALTH CONTRACT NO.: REAL PROPERTY							
		MARK	DATE	BY	DESCRIPTION		
		REVISIONS					
		U S D A			Agricultural Research Service		
US SALINITY LABORATORY UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSL							
MECHANICAL LOWER LEVEL OVERALL FLOOR PLAN							
			PROJECT NO:	USDA AGRICULTURAL RESEARCH SERVICE		SHEET ID	
		Project Number	USDA/IARS, PA		M100		
		ISSUE DATE:	800 BUCHANNAN STREET		SHT 9 OF 17		
		2-26-2025	ALBANY, CA				

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1 MECHANICAL MID LEVEL OVERALL FLOOR PLAN
1/16" = 1'-0"

KALARCHITECTS ARCHITECTURE ENGINEERING CONSTRUCTION MANAGEMENT A-E FIRM USDA PROJ. MANAGER: EPM Designer DRAWN BY: PPM Author CHECKED BY: SAFETY & HEALTH Checker CONTRACT NO.: REAL PROPERTY							
		MARK	DATE	BY	DESCRIPTION		
		REVISIONS					
		U S D A	 Agricultural Research Service US SALINITY LABORATORY UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSL MECHANICAL MID LEVEL OVERALL FLOOR PLAN				
		PROJECT NO:	USDA AGRICULTURAL RESEARCH SERVICE		SHEET ID		
		Project Number	USDA/ARS, PA		M101		
		ISSUE DATE:	800 BUCHANNAN STREET		SHT 10 OF 17		
		2-26-2025	ALBANY, CA				

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MECHANICAL FLOOR PLAN-MECHANICAL
YARD - DEMO

1/4" = 1'-0"

DEMOLISION KEYED NOTES

- REMOVE AND DEMOLISH EXISTING 200-TON CHILLER INCLUDING IING, WIRING, CONTROL INSTRUMENT AND ALL RELATED ACCESSORIES.
- POD TO EXISTING 6" CHILLED WATER SUPPLY AND RETURN.
- REMOVE AND DEMOLISH EXISTING WATER STORAGE TANK WITH ALL ACCESSORIES AND PIPING.
- REMOVE AND DEMOLISH WATER SOFTENER SYSTEM.
- REMOVE AND DEMOLISH ACTIVE CARBONE TANK.
- REMOVE AND DEMOLISH EXITING COOLING TOWER WITH ALL RELATED ACCESSORIES.
- REMOVE AND DEMOLISH EXISTING PRE COOLING WATER PUMPS WITH PIPING, WIRING, FITTING AND ALL RELATED ACCESSORIES.

MECHANICAL YARD FLOOR PLAN-NEW

1/4" = 1'-0"

RENOVATION KEYED NOTES

- PROVIDE NEW 132-TON CHILLER WITH BACNET CONTROL SYSTEM.
- POINT OF CONNECTION TO EXISTING 6" CHILLED WATER SUPPLY AND RETURN.
- ROVIDE A NEW 700-GALLON BUFFER TANK ON THE CHILLED WATER RETURN WITH ACCESSORIES.
- PROVIDE AND INSTALL A 6" BYPASS FOR BUFFER TANK.
- PROPORTIONAL ISOLATION VALVE TO PREVENT WATER FLOWING THROUGH A DISABLED CHILLER.
- MANUAL ISOLATION VALVE ON SUPPLY AND RETURN FOR MAINTENANCE.
- INSTALL FLEXIBLE JOINTS ON THE SUPPLY AND RETURN CONNECTIONS OF THE CHILLER TO ABSORB VIBRATIONS, ACCOMMODATE THERMAL EXPANSION, AND SIMPLIFY INSTALLATION AND REMOVAL OF THE CHILLER FOR MAINTENANCE OR REPLACEMENT BY ALLOWING SOME MOVEMENT IN THE CONNECTIONS.

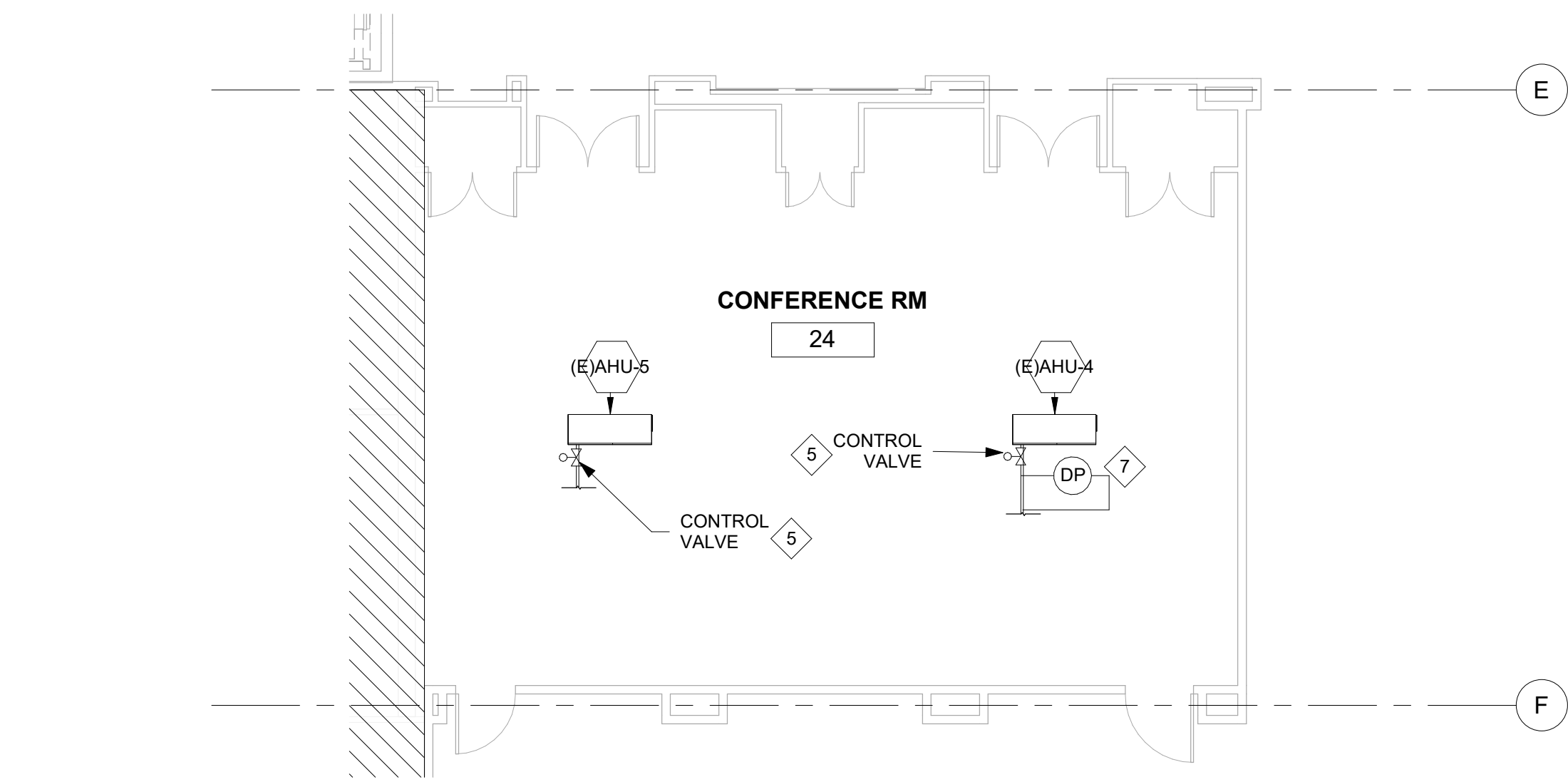


A-E FIRM	USDA
PROJ. MANAGER:	EPM
Designer	
DRAWN BY:	PPM
Author	
CHECKED BY:	SAFETY & HEALTH
Checker	
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REVISIONS			
U	Agricultural Research Service		
S			
D	US SALINITY LABORATORY		
A	UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA		
ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSL			
MECHANICAL YARD FLOOR PLAN			
PROJECT NO:	USDA AGRICULTURAL RESEARCH SERVICE	SHEET ID	
Project Number	USDA/ARS, PA	M201	
ISSUE DATE:	800 BUCHANNAN STREET		
2-26-2025	ALBANY, CA		SHT 11 OF 17

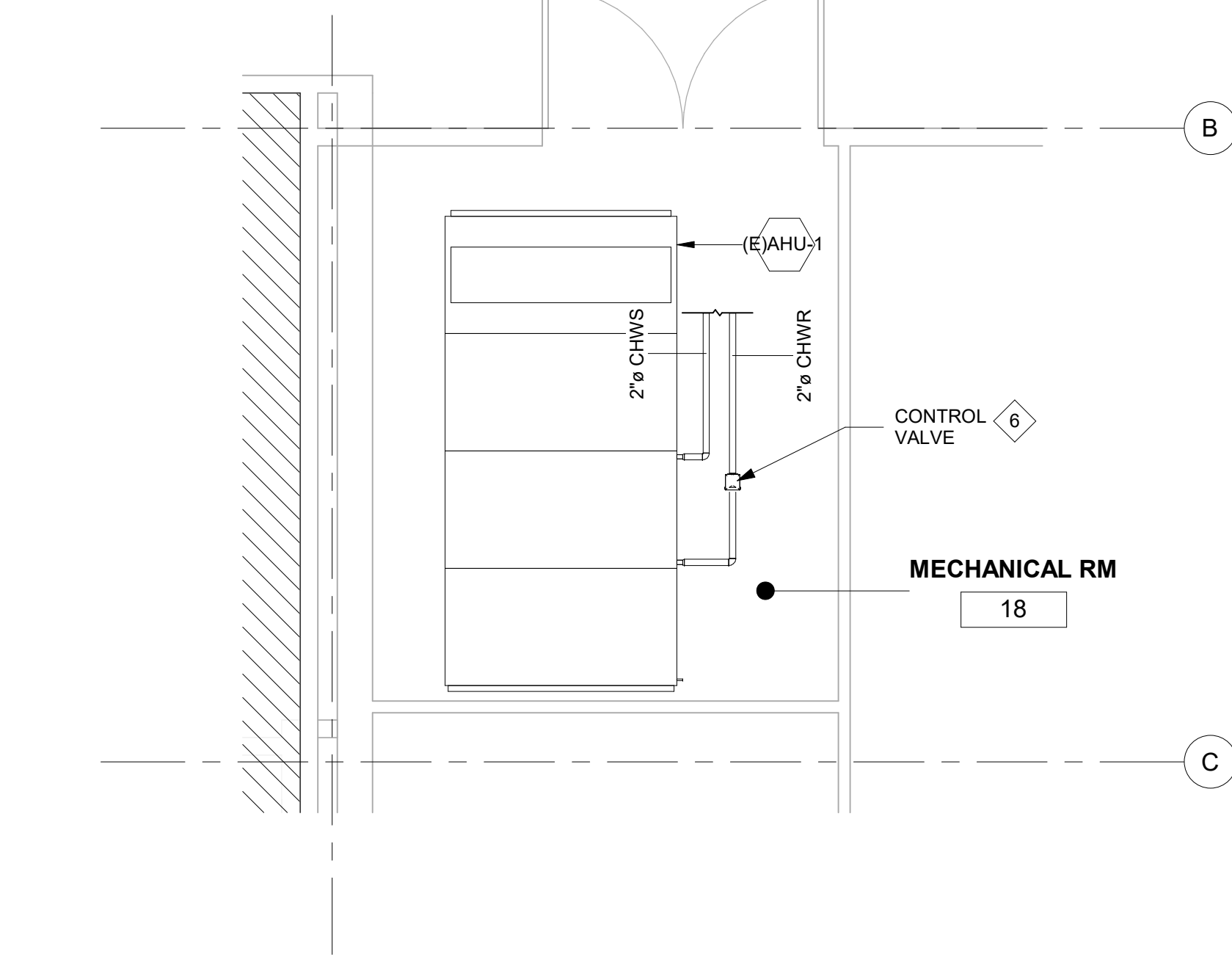
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Plot Date: 2/19/2025 11:18:00 PM

1 MECHANICAL FLOOR PLAN-MECHANICAL ROOM 126-DEMO
1/4" = 1'-0"



3 MECHANICAL FLOOR PLAN-CONFERENCE ROOM 24
1/8" = 1'-0"

2 MECHANICAL FLOOR PLAN-MECHANICAL ROOM 126-NEW
1/4" = 1'-0"



4 MECHANICAL FLOOR PLAN-MECHANICAL ROOM 18
1/4" = 1'-0"

DEMOLITION KEYED NOTES

- REMOVE AND DEMOLISH EXISTING CHILLED WATER PUMPS AND ALL RELATED ACCESSORIES..
- REMOVE AND DEMOLISH EXISTING CONTROL PANEL FOR CHILLED WATER PUMPS.
- KEEP AND PROTECT EXISTING VFD FOR NEW CHWS & CHWR.
- REMOVE AND DEMOLISH EXISTING BYPASS LINE AND 3-WAY VALVE.
- REMOVE EXISTING 3-WAY VALVE.
- EXISTING AIR SEPERATOR TO BE REMAIN.

RENOVATION KEYED NOTES

- PROVIDE NEW CHILLED WATER PUMPS FOR NEW CHILLER WITH ALL ACCESSORIES INCLUDING AIR SEPARATOR, FLEXIBLE CONNECTOR, GATE VALVE AND PRESSURE GAUGE.
- PROVIDE A NEW CONTROL PANEL FOR CHILLED WATER PUMPS.
- REPLACE 3-WAY VALVE WITH A 3", 2-WAY VALVE ON THE CHILLED WATER RETURN PIPE TO REGULATE FLOW AND MAINTAIN SYSTEM EFFICIENCY.
- REPLACE 3-WAY VALVE WITH A 6", 2-WAY VALVE ON THE CHILLED WATER RETURN PIPE TO REGULATE FLOW AND MAINTAIN SYSTEM EFFICIENCY.
- REPLACE 3-WAY VALVE WITH A 1-1/2", 2-WAY VALVE ON THE CHILLED WATER RETURN PIPE TO REGULATE FLOW AND MAINTAIN SYSTEM EFFICIENCY.
- REPLACE 3-WAY VALVE WITH A 2", 2-WAY VALVE ON THE CHILLED WATER RETURN PIPE TO REGULATE FLOW AND MAINTAIN SYSTEM EFFICIENCY.
- PROVIDE DIFFERENTIAL PRESSURE (DP) AT THE END OF THE BUILDING'S CHILLED WATER LOOP.
- THE PUMP SPEEDS ARE MODULATED TO MAINTAIN THE DIFFERENTIAL SET POINT.
- INSTALL A NEW 6" BYPASS WITH MANUAL VALVE TO ISOLATE THE SYSTEM FROM CHILLERS AND BUFFER TANK DURING COMMISSIONING OR MAINTENANCE.
- PROVIDE AND INSTALL A 3" BYPASS LINE IN THE CHILLED WATER SYSTEM WITH AN AUTOMATIC MODULATING VALVE TO MAINTAIN THE MINIMUM REQUIRED FLOW RATE FOR CHILLER PROTECTION AND PERFORMANCE.



ARCHITECTURE|ENGINEERING|CONSTRUCTION MANAGEMENT

A-E FIRM

PROJ. MANAGER: SZ

DRAWN BY: SZ

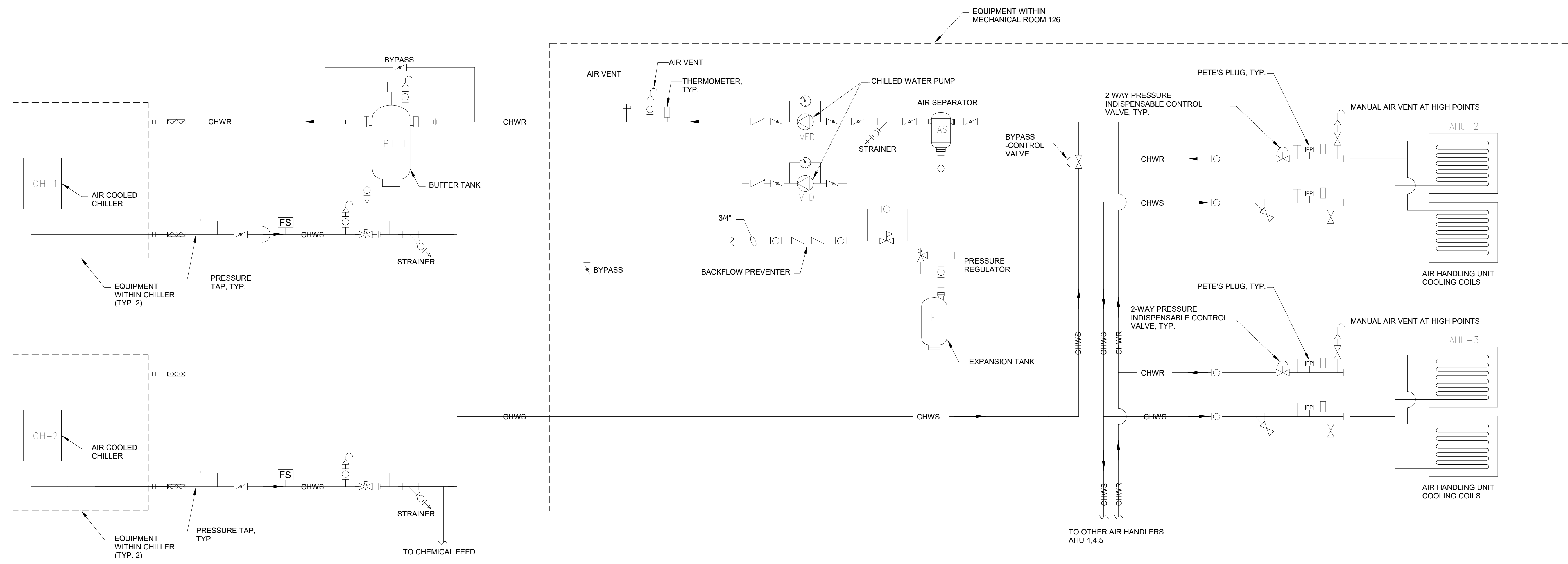
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CONTRACT NO.: REAL PROPERTY

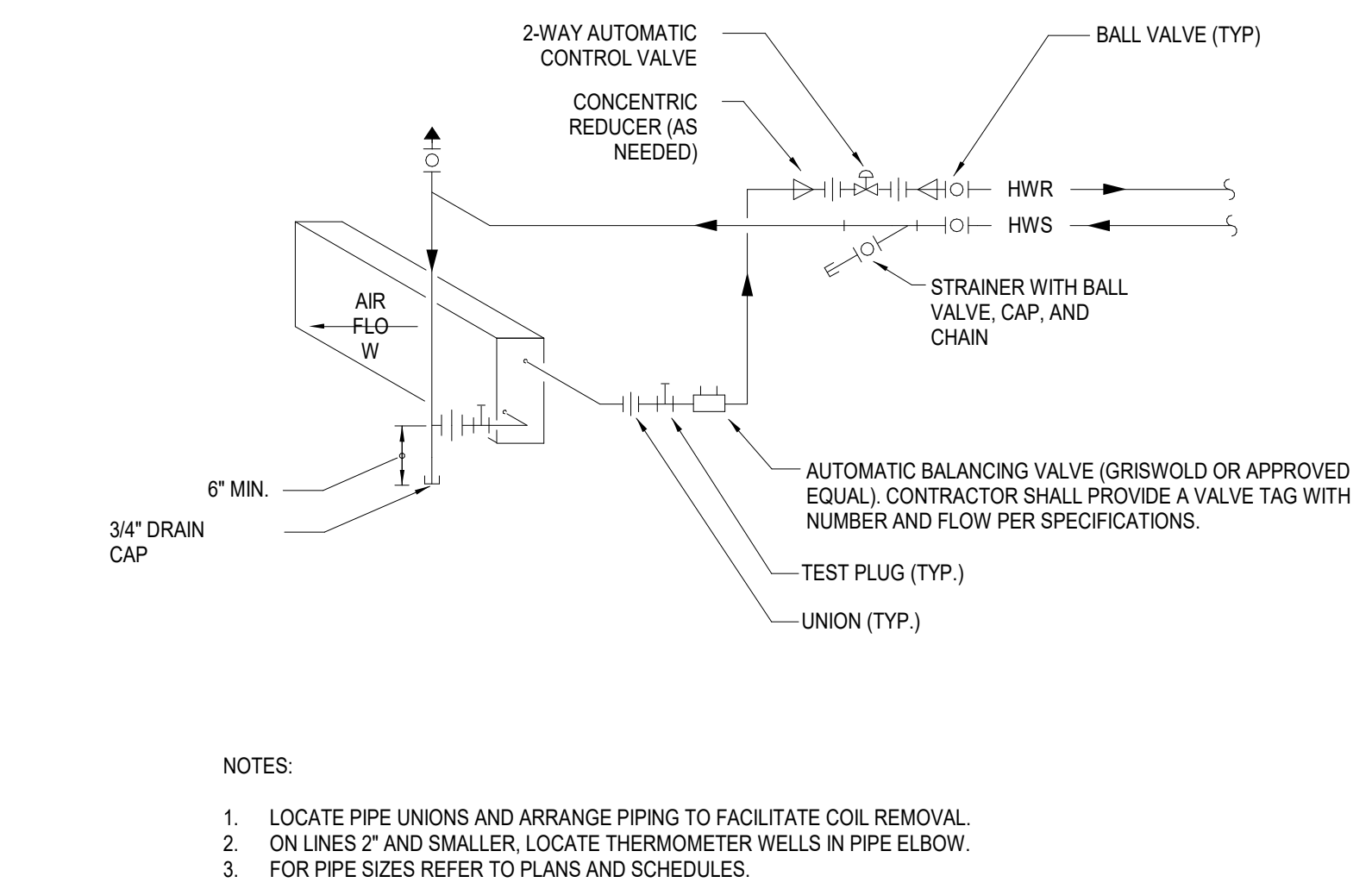
MARK	DATE	BY	DESCRIPTION
U S D A	REVISIONS		
	 Agricultural Research Service		
	US SALINITY LABORATORY		
	UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA		
	ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSR		
A	MECHANICAL ENLARGED PLAN		
#	PROJECT NO: Project Number	USDA AGRICULTURAL RESEARCH SERVICE USDA/ARS, PA	SHEET ID
	ISSUE DATE: 2-26-2025	800 BUCHANNAN STREET ALBANY, CA	M202
			SHT 12 OF 17

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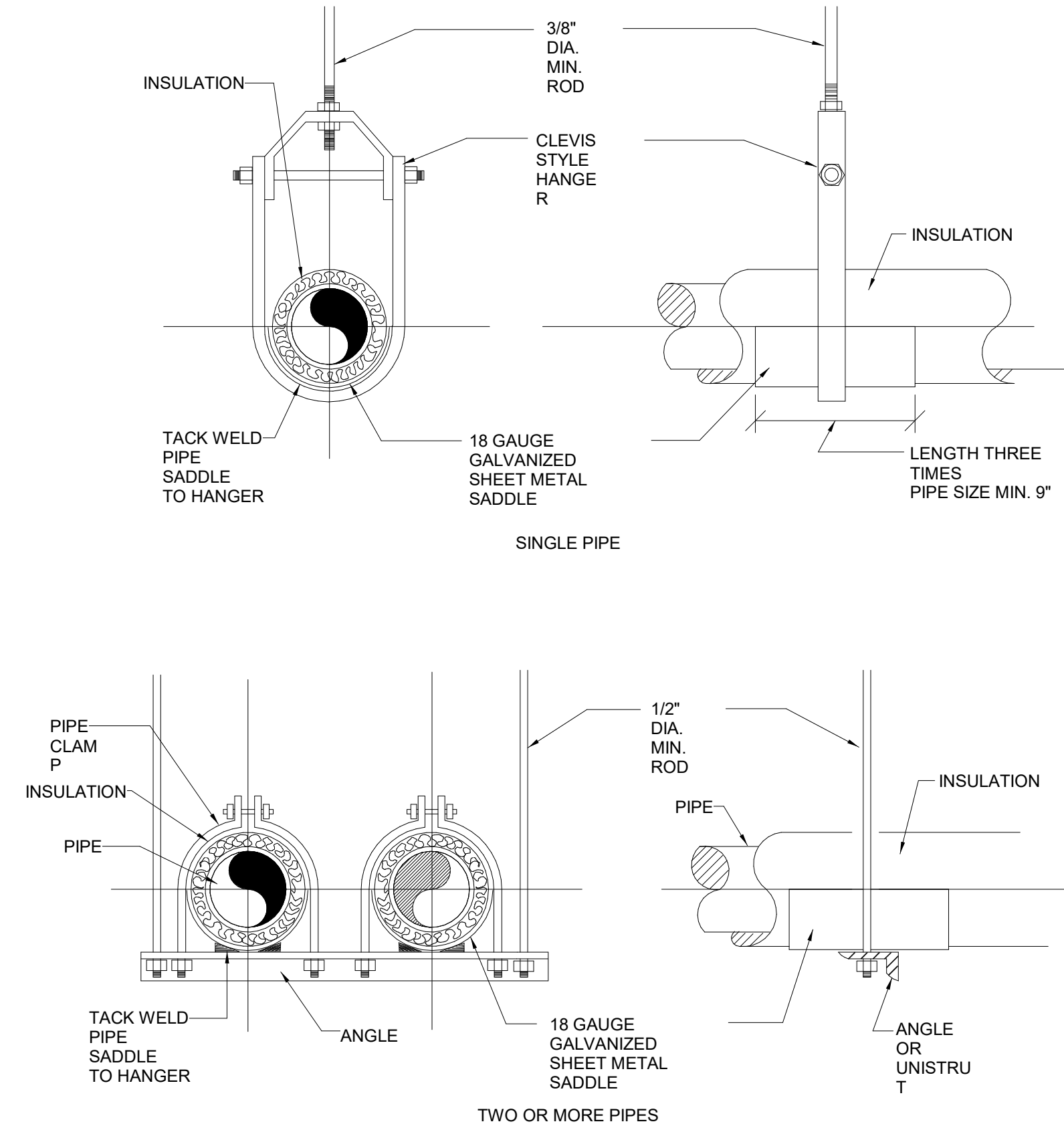
1 CHILLER SYSTEM PIPING DIAGRAM
1" = 1'-0"



2 Cooling Coil
1" = 1'-0"



3 Pipe Hanger
1" = 1'-0"



STEEL ROD SIZING SCHEDULE	
PIPE, IPS	ROD
TO 2 IN	3/8 IN
2-1/2 IN	1/2 IN
3 IN	1/2 IN
4 IN	5/8 IN
6 IN	3/4 IN

PIPE SUPPORT SPACING SCHEDULE		
TYPE OF PIPE	SIZE	MAX. SPACING
STEEL	1-1/2" AND SMALLER	7 FT
	2" AND LARGER	10 FT
BRASS OR COPPER	3/4" AND SMALLER	5 FT
	1" TO 1-1/4"	6 FT
	1-1/2" TO 3"	8 FT
	4" AND LARGER	10 FT

ARCHITECTURE|ENGINEERING|CONSTRUCTION MANAGEMENT

PROJ. MANAGER:

Designer

DRAWN BY:

Author

CHECKED BY:

Checker

CONTRACT NO.:

REAL PROPERTY

USDA

Agricultural Research Service

US SALINITY LABORATORY

UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA

ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSL

MECHANICAL DETAIL

MARK

DATE

BY

DESCRIPTION

U

S

D

A

PROJECT NO:

Project Number

ISSUE DATE:

2-26-2025

USDA AGRICULTURAL RESEARCH SERVICE

USDA/IARS, PA

800 BUCHANNAN STREET

ALBANY, CA

SHEET ID

M300

SHT 13 OF 17

Autodesk Docs (USDA) TO-02 (Revised) Elevator & Chiller Replacement (USDA) USSL - E01.rvt
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D																							
C																							
B																							
A																							

ABBREVIATIONS:

#	NUMBER	INF	INFORMATION
AC	ALTERNATING CURRENT	ISP	INSIDE PLANT
ADN	AREA DISTRIBUTION NODE	ITE	INFORMATION TECHNOLOGY EQUIPMENT
AFF	ABOVE FINISHED FLOOR	JB	JUNCTION BOX
AFG	ABOVE FINISHED GRADE	KVA	KILOVOLT AMPS
AHU	AIR HANDLER UNIT	LCP	LIGHTING CONTROL PANEL
AIC	AMPS INTERRUPTING CURRENT	LED	LIGHT EMITTING DIODE
ALIS	AUTONOMIC LOGISTICS INFORMATION SYSTEM	LV	LOW VOLTAGE
ATS	AUTOMATIC TRANSFER SWITCH	MC	METAL CLAD
AUTO	AUTOMATIC	MM	MILLIMETER
AV	AUDIO VISUAL	MSB	MASTER SWITCHBOARD
AWG	AMERICAN WIRE GAUGE	MTD	MOUNTED
BFC	BELOW FINISHED CEILING	(N)	NEW
BFF	BELOW FINISHED FLOOR	NEC	NATIONAL ELECTRICAL CODE
BFG	BELOW FINISHED GRADE	NEMA	NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
C	CONDUIT	NFPA	NATIONAL FIRE PREVENTION ASSOCIATION
CEC	CALIFORNIA ELECTRICAL CODE	N.I.E.S.	NOT IN ELECTRICAL SCOPE
CCTV	CLOSED CIRCUIT TELEVISION	NL	NIGHT LIGHT
CAT	CATEGORY	N.O.	NORMALLY OPEN
CH	CHILLER	NTS	NOT TO SCALE
CHWP	CHILLED WATER PUMP	OTS	OPEN TO STRUCTURE
CKT	CIRCUIT	PH, ∅	PHASE
CO	CONDUIT ONLY	PA	PUBLIC ADDRESS SYSTEM
COMM	COMMUNICATIONS (TELECOMM)	PB	PULL BOX
CRAC	COMPUTER ROOM AIR CONDITIONER	PDU	POWER DISTRIBUTION UNIT
CRI	COLOR RENDITION INDEX	POP	POST OFFICE PROTOCOL
CT	CONDENSER TOWER	(R)	REMOVE
CT	CURRENT TRANSFORMER	RAH	RETURN AIR HANDLER
CU	COPPER	RF	RADIO FREQUENCY
CWP	CONDENSER WATER PUMP	RFP	REQUEST FOR PROPOSAL
DC	DIRECT CURRENT	RM	ROOM
DISC	DISCONNECT	RSC	RIGID STEEL CONDUIT
(E)	EXISTING	SAH	SUPPLY AIR HANDLER
ECC	EMERGENCY COMMUNICATIONS CENTER	SAN	STORAGE AREA NETWORK
EF	EXHAUST FAN	SEC	SECURITY
ELEC	ELECTRICAL	SCE	SOUTHERN CALIFORNIA EDISON (UTILITY COMPANY)
EM	EMERGENCY	SF	SUPPLY FAN
EMT	ELECTRICAL METALLIC TUBING	SFD	SMOKE/FIRE DAMPER
EPO	EMERGENCY POWER OFF	SIPR	SECRETE INFORMATION PROTOCOL ROUTER
EVAC	FIRE ALARM EVACUATION CONTROL	SM	SINGLE MODE
EVC	EVAPORATIVE COOLER	SMACNA	SHEET METAL & AIR CONDITIONING NATIONAL ASSOC.
(F)	FUTURE	SPD	SURGE PROTECTIVE DEVICE /
FA	FIRE ALARM	TEL	TELEPHONE
FACP	FIRE ALARM CONTROL PANEL	THRU	THROUGH
FATC	FIRE ALARM TERMINATION CABINET	TMGB	TELECOM MAIN GROUND BUS
FLA	FULL LOAD AMPS	TTB	TELECOM TERMINAL BOARD
FLEX	FLEXIBLE METAL CONDUIT	TX	TRANSFORMER
FLMC	LIQUIDTIGHT METALLIC CONDUIT	TVSS	TRANSIENT VOLTAGE SURGE SUPPRESSION
FO	FIBER OPTIC	(TYP.)	TYPICAL
G, GND	GROUND	UFC	UNIFORM FIRE CODE
GA	GAUGE	UG	A CONCRETE ENCASED GROUND ELECTRODE METHOD
GFCI	GROUND FAULT CIRCUIT INTERRUPTER	UFER	UNDERGROUND
GRC	GALVANIZED RIGID CONDUIT	UL	UNDERWRITERS LABORATORY
HP	HORSE POWER	U.O.N.	UNLESS OTHERWISE NOTED
HID	HIGH INTENSITY DISCHARGE	UPS	UNINTERRUPTIBLE POWER SUPPLY
HVAC	HEATING, VENTILATING & AIR CONDITIONING	V	VOLT
HWP	HOT WATER PUMP	VA	VOLTAMPERES
IDF	INTERMEDIATE DISTRIBUTION FRAME	VDC	VOLTS DIRECT CURRENT
IES	INTERNATIONAL ELECTROTECHNICAL STANDARDS	WP	WEATHERPROOF
IG	ISOLATED GROUND	XFMR	TRANSFORMER
		XP	EXPLOSIONPROOF

GENERAL NOTES:

- THE ELECTRICAL SPECIFICATIONS ARE AN INTEGRAL PART OF THE ELECTRICAL CONSTRUCTION DOCUMENTS. THE CONTRACTOR SHALL THOROUGHLY REVIEW ALL INFORMATION PROVIDED IN BOTH THESE DRAWINGS AND THE ASSOCIATED ELECTRICAL SPECIFICATIONS PRIOR TO PREPARING A BID FOR THE ELECTRICAL WORK AND SHALL AFFIRM HAVING DONE SO BY THE SUBMISSION OF A BID FOR STATED WORK.
- FURNISH AND INSTALL ALL NECESSARY LABOR, MATERIALS, EQUIPMENT AND INCIDENTALS REQUIRED TO INSTALL COMPLETE AND OPERATIONAL ELECTRICAL SYSTEMS ACCORDING TO THE INTENT OF THESE DRAWINGS AND ASSOCIATED SPECIFICATIONS WHETHER ITEMIZED OR NOT.
- EXAMINE THE DRAWINGS FOR MECHANICAL EQUIPMENT AND PROVIDE STARTERS, CIRCUIT BREAKERS, SWITCHES, PUSH BUTTONS AND AUXILIARY DEVICES AS REQUIRED. MAKE ALL ELECTRICAL CONNECTIONS TO PLACE THE EQUIPMENT IN COMPLETE OPERATION.
- THE CONTRACTOR SHALL HAVE THOROUGHLY EXAMINED THE SITE AND FAMILIARIZED HIMSELF WITH THE EXISTING CONDITIONS, AND SHALL HAVE MADE ALLOWANCE THEREFORE IN PREPARING HIS PROPOSAL. THE ELECTRICAL DRAWINGS ARE DIAGRAMATIC AND THE LOCATION OF THE ELECTRICAL DEVICES SHALL BE COORDINATED WITH THE ARCHITECT.
- IN THE EVENT OF DISCREPANCIES BETWEEN SPECIFICATIONS AND THE DRAWINGS, THE ELECTRICAL CONTRACTOR SHALL BID NEW CONDITIONS, WIRES AND NECESSARY EQUIPMENT IN ORDER TO COMPLETE THE JOB AND PROVIDE A FULLY OPERABLE AND ACCEPTABLE SYSTEM. EXTRAS WILL NOT BE ALLOWED FOR WORK NOT INDICATED OR NOTED ON THE DRAWINGS.
- THE CODE REFERENCES IN THESE GENERAL NOTES ARE "MINIMUM REQUIREMENTS" AND ALL ORK SHALL BE IN COMPLIANCE WITH CURRENT EDITIONS OF OSHA, NFPA 70, NFPA 70E, AND ALL LOCAL CODES.
- FIRE SEAL AROUND ALL CONDUIT PENETRATIONS THROUGH FIRE BARRIERS WITH AN APPROVED FIRE SEALANT EQUAL TO THE RATING OF THE SURFACE PENETRATED. FIRE SEAL INSIDE OF CONDUIT AFTER CONDUCTOR INSTALLATION.

SCOPE OF WORK

- REMOVE EXISTING FEEDER TO EXISTING CHILLER TO BE DEMOLISHED.
- REMOVE EXISTING DISCONNECT SWITCHES SERVING EXISTING CHILLER, PUMPS, & COOLING TOWER TO BE DEMOLISHED.
- PROVIDE ELECTRICAL CONNECTION TO NEW CHILLERS.
- PROVIDE NEW CIRCUIT BREAKERS IN EXISTING MAIN SWITCHBOARD TO SERVE NEW CHILLERS.
- DISCONNECT ELECTRICAL CONNECTION TO PUMP MOTOR CONTROLLER AND PROVIDE NEW CONNECTION TO NEW CONTROLLER.
- PROVIDE COORDINATION STUDY TO NEW CIRCUIT BREAKERS.
- EXTEND EXISTING 120V POWER CONNECTION TO NEW CHILLERS AS REQUIRE.

ELECTRICAL LEGEND:

	LED RECESSED 2'x4' LIGHT FIXTURE ON T-BAR CEILING. REFER TO LUMINAIRE SCHEDULE.
NL	EMERGENCY LED LIGHT FIXTURE CONNECTED TO EMERGENCY LIGHTING INVERTER. NL=NIGHT LIGHT, ALWAYS ON.
	1'x4' LED LIGHT FIXTURE. REFER TO LUMINAIRE SCHEDULE.
	4' LED STRIP LIGHT FIXTURE. REFER TO LUMINAIRE SCHEDULE.
	LED DOWN LIGHT FIXTURE. REFER TO LUMINAIRE SCHEDULE.
	EXISTING 2'x4' LIGHT FIXTURE TO BE REMOVED.
	EXISTING DOWN LIGHT FIXTURE TO BE REMOVED.
	EXISTING STRIP LIGHT FIXTURE TO BE REMOVED.
	WALL MOUNTED LIGHT SCONCE, INTERIOR OR EXTERIOR REFER TO LUMINAIRE SCHEDULE
	WALL MOUNTED EXTERIOR LIGHT FIXTURE REFER TO LUMINAIRE SCHEDULE
	WALL MOUNTED EXTERIOR LIGHT FIXTURE CONNECTED TO EMERGENCY LIGHTING INVERTER
A,A1	LUMINAIRE FIXTURE TAG: LETTER INDICATES TYPE
a,b,c	INDICATES LUMINAIRE DIMING ZONE/ DESIGNATION
	TWIN WALL MTD. EMERGENCY LUMINAIRE WITH 90 MINUTES BATTERY PACK
	FLUORESCENT / HID LIGHT FIXTURE - RECESSED OR SURFACE MOUNTED
	FLUORESCENT / HID LIGHT FIXTURE - WALL MOUNTED
	EXIT LIGHT - CEILING MOUNTED WITH ARROWS AS SHOWN
	EXIT LIGHT - WALL MOUNTED WITH ARROWS AS SHOWN
	SINGLE POLE TOGGLE SWITCH, @ +48" MAX TOP OF BOX UON
	THREE-WAY TOGGLE SWITCH, @ +48" MAX TOP OF BOX UON
	LINE VOLTAGE DIMMER, @ +48" MAX TOP OF BOX UON
	MANUAL STARTER, FRACTIONAL HORSEPOWER MOTOR
	LOW VOLTAGE, WALL MOUNTED DIMMER SWITCH, @ +48" MAX TOP OF BOX UON, LOWER CASE LETTER INDICATES DIMMING ZONE.
	LIGHTING ROOM CONTROLLER OR SMART PACK, MOUNTED ABOVE ACCESSIBLE CEILING OR WALL MOUNTED @ +48" MAX TOP OF BOX AS NOTED, LOWER CASE LETTER INDICATES DIMMING ZONE.
	PHOTOCELL
	JUNCTION BOX, SIZE & TYPE AS INDICATED OR AS REQUIRED
	SPECIAL RECEPTACLE AS NOTED, @ +18" UON
	20 AMP 125V 3W DUPLEX RECEPTACLE, @ +18" UON
	20 AMP 125V 3W DUPLEX RECEPTACLE, HALF SWITCHED, @ +18" UON
	20 AMP 125V 3W DUPLEX RECEPTACLE, WITH IG (ISO GND), COLUMN MOUNTED
	20 AMP 125V 3W DUPLEX RECEPTACLE WITH AFCI (ARC FAULT CIRCUIT INTERRUPTER) TYPE, @ +18" UON
	20 AMP 125V 3W QUAD RECEPTACLE WITH AFCI (ARC FAULT CIRCUIT INTERRUPTER) TYPE, @ +18" UON
	20 AMP 125V 3W QUAD RECEPTACLE, @ +18" UON
	20 AMP 125V 3W QUAD RECEPTACLE, (1) DUPLEX SWITCHED, @ +18" UON
	20 AMP 120V DUPLEX RECEPTACLE, WITH GFCI, @ +44" UON
	20 AMP 120V DUPLEX RECEPTACLE, WEATHER-PROOF WITH GFCI, @ +44" UON
	20 AMP 125V 3W QUAD RECEPTACLE, WITH GFCI @ +44" UON
	30 AMP 208V 3W RECEPTACLE
	COMBINATION BREAKER/STARTER, SIZE PER UNIT LABEL
	NON-FUSED DISCONNECT SWITCH, SIZE PER UNIT LABEL
	FUSED DISCONNECT SWITCH, SIZE PER UNIT LABEL
	EQUIPMENT STARTER, SIZE PER UNIT LABEL

	MOTOR, N.I.E.S. CONNECT AS REQUIRED, NUMBER INDICATES HP
	PANELBOARD - SURFACE MOUNTED - SEE SCHEDULE
	PANELBOARD - FLUSH MOUNTED - SEE SCHEDULE
	MAIN SWITCHBOARD OR MOTOR CONTROL CENTER, SEE ONE LINE DIAGRAM
	CONDUIT CONCEALED IN CEILING OR WALL. BRANCH CIRCUIT WITHOUT FURTHER DESIGNATION INDICATES #12 AWG WIRE CIRCUIT AND 3/4" CONDUIT. ALL CONDUITS AND RACEWAY MUST HAVE AN INSULATED GROUND WIRE SIZED PER NEC 250.122.
	WIRE 'TICK MARKS' INDICATE: 2 #12 AWG + 1 #12 NEUTRAL + GND, UON. (GROUND WIRE NOT SHOWN, BUT REQUIRED FOR ALL CIRCUITS) HOMERUN TO RESPECTIVE PANEL OR TERMINAL CABINET - OVERHEAD
	HOMERUN TO RESPECTIVE PANEL OR TERMINAL CABINET - UNDERGROUND
	CONDUIT RISER - UP
	CONDUIT RISER - DOWN
	FLEXIBLE CONDUIT CONECTION
	CONDUIT RUN CONCEALED BELOW FLOOR OR FINISHED GRADE BRANCH CIRCUIT WITHOUT FURTHER DESIGNATION INDICATES #12 AWG WIRE CIRCUIT AND 3/4" CONDUIT. ALL CONDUITS AND RACEWAY MUST HAVE AN INSULATED GROUND WIRE SIZED PER NEC 250.122.
	INDICATES MECHANICAL, OR PLUMBING, EQUIPMENT DESIGNATION TAG. SEE MECHANICAL AND PLUMBING SCHEDULES FOR ADDITIONAL INFO. (EXAMPLE: EF-1 = EXHAUST FAN, REFER TO MECHANICAL EQUIPMENT SCHEDULE.)
	SECTION DESIGNATION; TOP LETTER INDICATES SECTION, BOTTOM LETTER/NUMBER INDICATES SHEET
	DETAIL DESIGNATION; TOP NUMBER INDICATES DETAIL, BOTTOM LETTER/NUMBER INDICATES SHEET
	KEY NOTE REFERENCE TAG.
	GROUND ROD
	TESTING GROUND WELL WITH GROUND ROD
	EXOTHERMIC WELD CONNECTION
	TRANSFORMER, MOUNTING AS NOTED

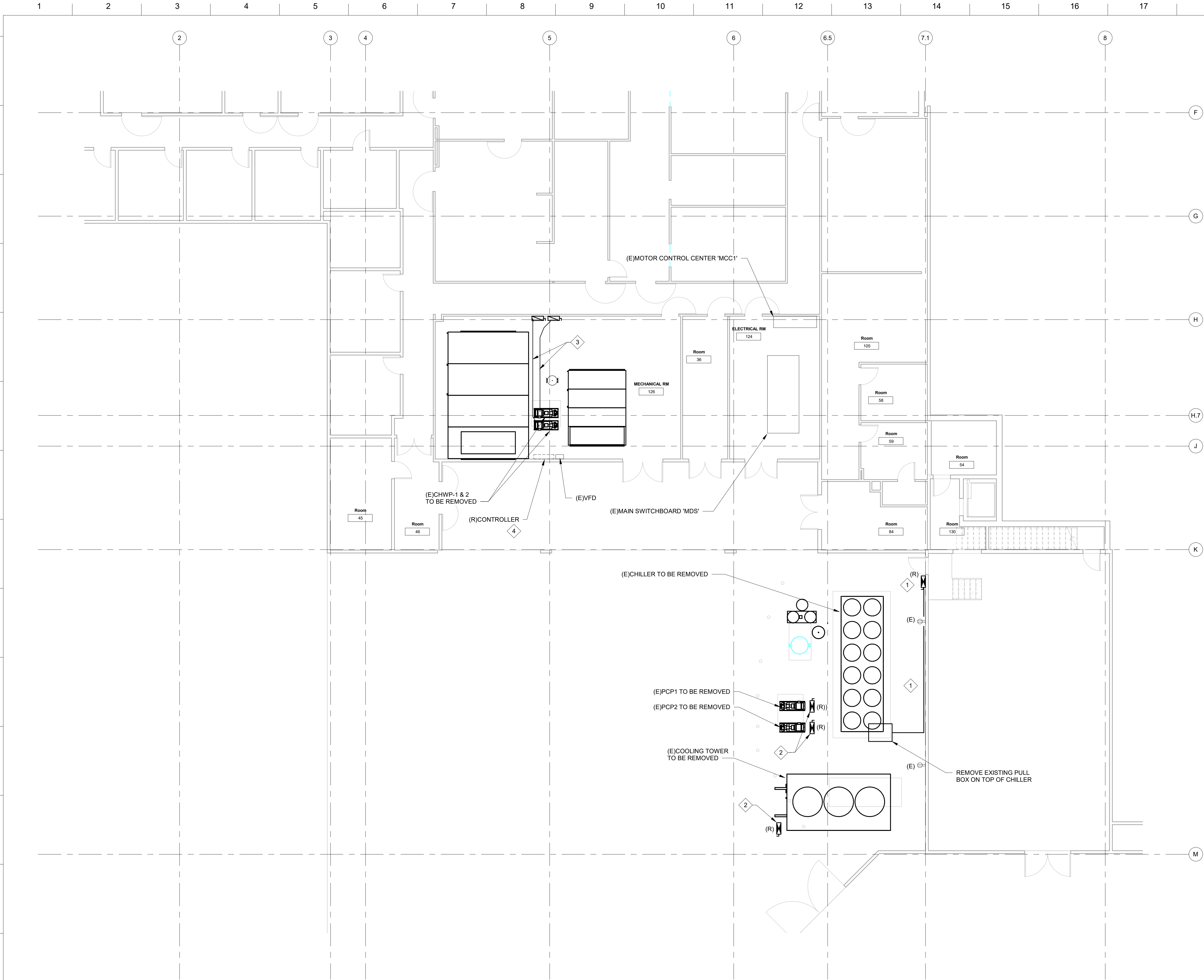
SINGLE LINE LEGEND:

	INDICATES ELECTRICAL FEEDER SIZE. REFER TO FEEDER SCHEDULE.
100A	CIRCUIT BREAKER. AMP SIZE AND POLES, AS INDICATED
	CIRCUIT SWITCH, SINGLE BREAK
	FUSED BREAKER
	TERMINAL(S)
	DIESEL GENERATOR
	AUTOMATIC TRANSFER SWITCH
	ELECTRICAL METER WITH CURRENT TRANSFORMERS
	POTENTIAL TRANSFORMER
	FUSE
	VOLTAGE TRANSFORMER
	EARTH GROUND

NOTE:
SYMBOLS INDICATED ABOVE MAY NOT NECESSARILY APPEAR AS PART OF THESE DRAWINGS IF NOT REQUIRED.

KALARCHITECTS ARCHITECTURE/ENGINEERING/CONSTRUCTION MANAGEMENT					
		MARK	DATE	BY	DESCRIPTION
		REVISIONS			
A-E FIRM	USDA	U S D A	 Agricultural Research Service US SALINITY LABORATORY UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSL		
PROJ. MANAGER:	EPM		GENERAL NOTES, LEGEND, AND ABBREVIATIONS		
Designer					
DRAWN BY:	PPM				
Author					
CHECKED BY:	SAFETY & HEALTH	PROJECT NO:	USDA AGRICULTURAL RESEARCH SERVICE	SHEET ID	
Checker		Project Number	USDA/ARS, PA	E001	
CONTRACT NO.:	REAL PROPERTY	ISSUE DATE:	800 BUCHANNAN STREET	SHT 14 OF 17	
		2-26-2025	ALBANY, CA		

Autodesk Docs: USDA 10-02 Riverside Elevator & Chiller Replacement\USDA\USSL_E100.rvt
File Path:
2/26/2025 8:38:12 AM
Plot Date:



1 ELECTRICAL DEMOLITION PARTIAL
FLOOR PLAN
1/8" = 1'-0"

KEY NOTES

1. DISCONNECT AND REMOVE EXISTING 600A DISCONNECT SWITCH SERVING EXISTING CHILLER TO BE REMOVED. REMOVE SURFACE MOUNTED FEEDER FROM DISCONNECT SWITCH TO CHILLER. REMOVE CONDUCTORS BACK TO SOURCE. REMOVE 120V FEEDER BACK TO EXISTING RECEPTACLE ON WALL. REMOVE CONTROL CONDUIT AND CABLES BACK TO BUILDING PENETRATION. CAP CONDUIT.
2. DISCONNECT AND REMOVE EXISTING DISCONNECT SWITCHES SERVING EXISTING PUMPS AND COOLING TOWER TO BE REMOVED. REMOVE FEEDER ABOVE GRADE. ABANDON CONDUITS BELOW GRADE. REMOVE CONDUCTORS BACK TO SOURCE. CAP AND LABEL EXISTING CONDUIT STUP-UP.
3. DISCONNECT AND REMOVE EXITING FEEDERS SERVING PUMPS TO BE REMOVE. BACK TO EXISTING DISCONNECT SWITCH. EXISTING DISCONNECT SWITCHES TO REMAIN.
4. DISCONNECT ELECTRICAL CONNECTION TO EXISTING PUMP CONTROLLER TO BE REMOVED. RETAIN CIRCUIT FOR NEW CONTROLLER.

KALARCHITECTS ARCHITECTURE ENGINEERING CONSTRUCTION MANAGEMENT A-E FIRM USDA PROJ. MANAGER: EPM DRAWN BY: QL CHECKED BY: Checker CONTRACT NO.: REAL PROPERTY					
	MARK	DATE	BY	DESCRIPTION	
	REVISIONS				
	U S D A			Agricultural Research Service	
				US SALINITY LABORATORY	
		UNIVERSITY OF CALIFORNIA, RIVERSIDE, CA			
		ELEVATOR MODERNIZATION AND CHILLER REPLACEMENT AT USSL			
ELECTRICAL DEMOLITION PARTIAL FLOOR PLAN					

PROJECT NO:	USDA AGRICULTURAL RESEARCH SERVICE	SHEET ID
Project Number	USDA/ARS, PA	E100
ISSUE DATE:	800 BUCHANNAN STREET	SHT 15 OF 17
2-26-2025	ALBANY, CA	

